

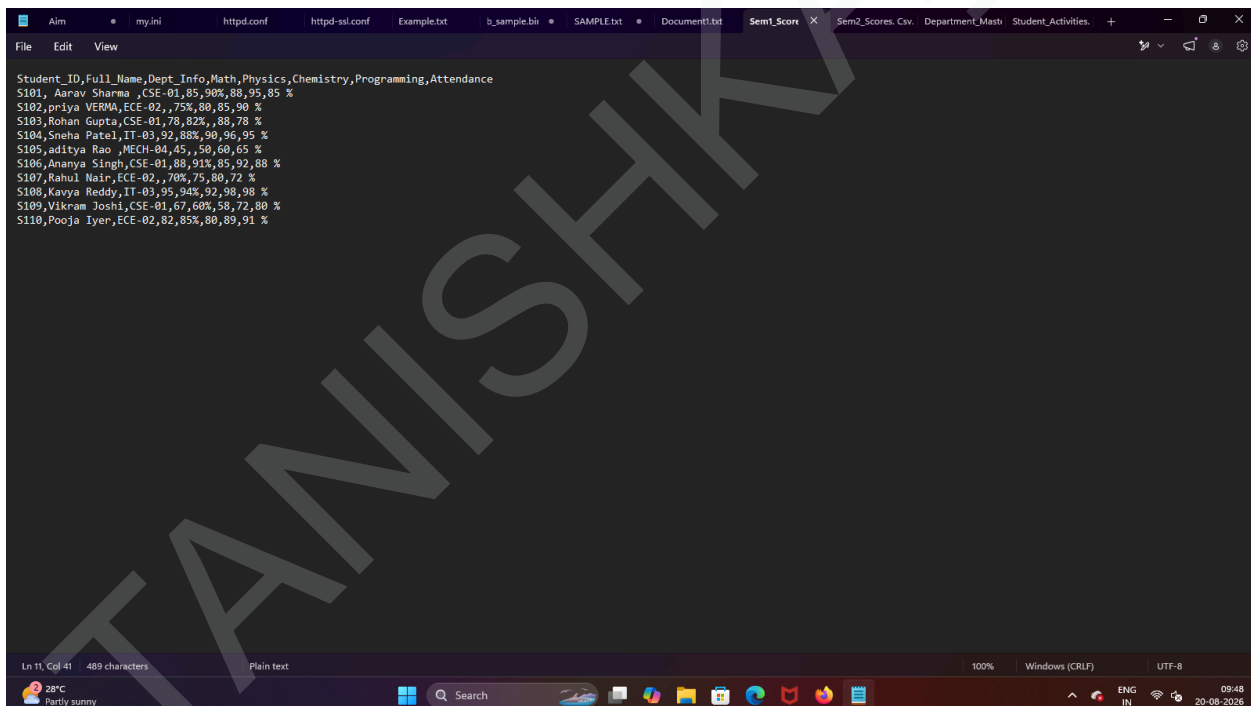
SHETH L.U.J AND SIR M.V COLLEGE
SUBJECT: - TABLEAU AND POWER BI
PRACTICAL NO: T089

AIM: Cleaning and Structuring Data for Analysis

- a) Identify and handle missing or inconsistent data
- b) Convert wide data into tall format
- c) Perform union of multiple files
- d) Execute cross-database joins
- e) Restructure data for better visualization
- f) Handle different levels of detail in datasets
- g) Apply advanced data cleaning techniques

create different csv files using Notepad/text Editor. Copy paste the data in Notepad/text Editor and save the files with .csv extension

1) Sem1_Scores.csv

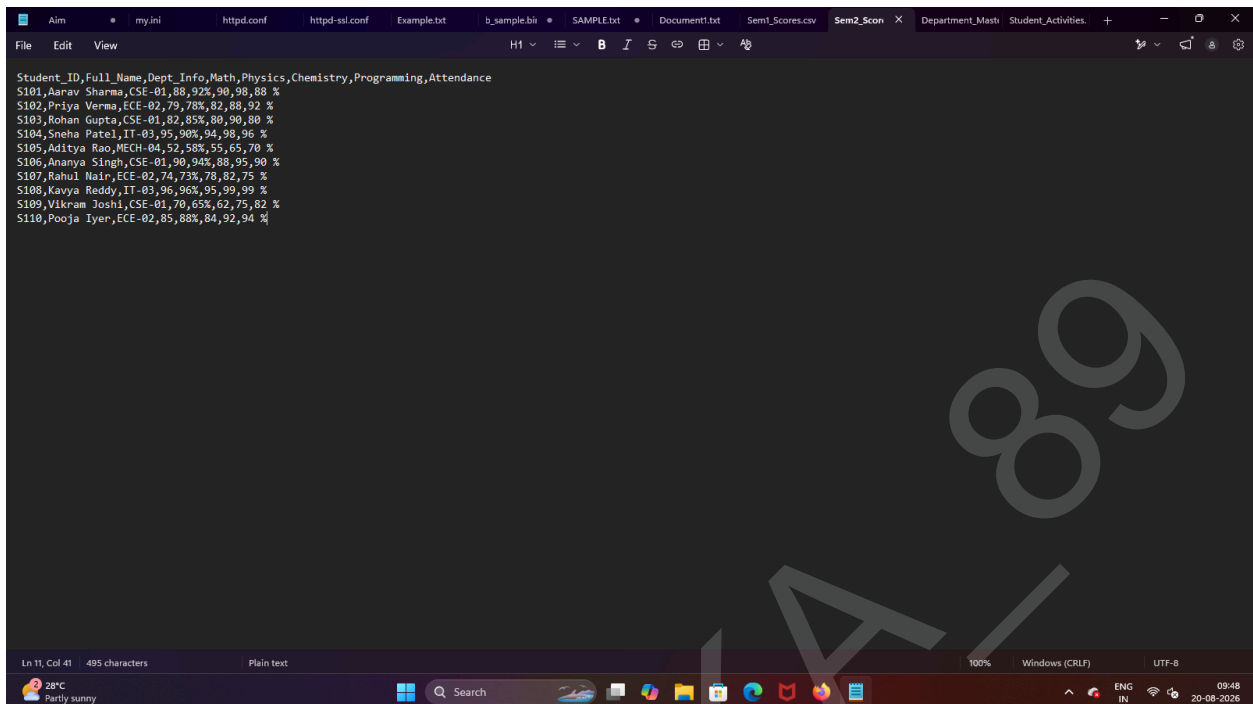


```
Student_ID,Full_Name,Dept_Info,Math,Physics,Chemistry,Programming,Attendance
S101,Aarav Sharma,CSE-01,85,90%,88,95,85 %
S102,Priya VERMA,ECE-02,,75%,88,85,90 %
S103,Rohan Gupta,CSE-01,78,82%,88,78 %
S104,Sneha Patel,IT-03,92,88%,90,96,95 %
S105,Aditya Rao,MECH-04,45,,50,60,65 %
S106,Ananya Singh,CSE-01,88,91%,85,92,88 %
S107,Rahul Nair,ECE-02,,70%,75,80,72 %
S108,Kavya Reddy,IT-03,95,94%,92,90,98 %
S109,Vikram Joshi,CSE-01,67,60%,58,72,80 %
S110,Pooja Iyer,ECE-02,82,85%,80,89,91 %
```

2) Sem2_Scores.Csv

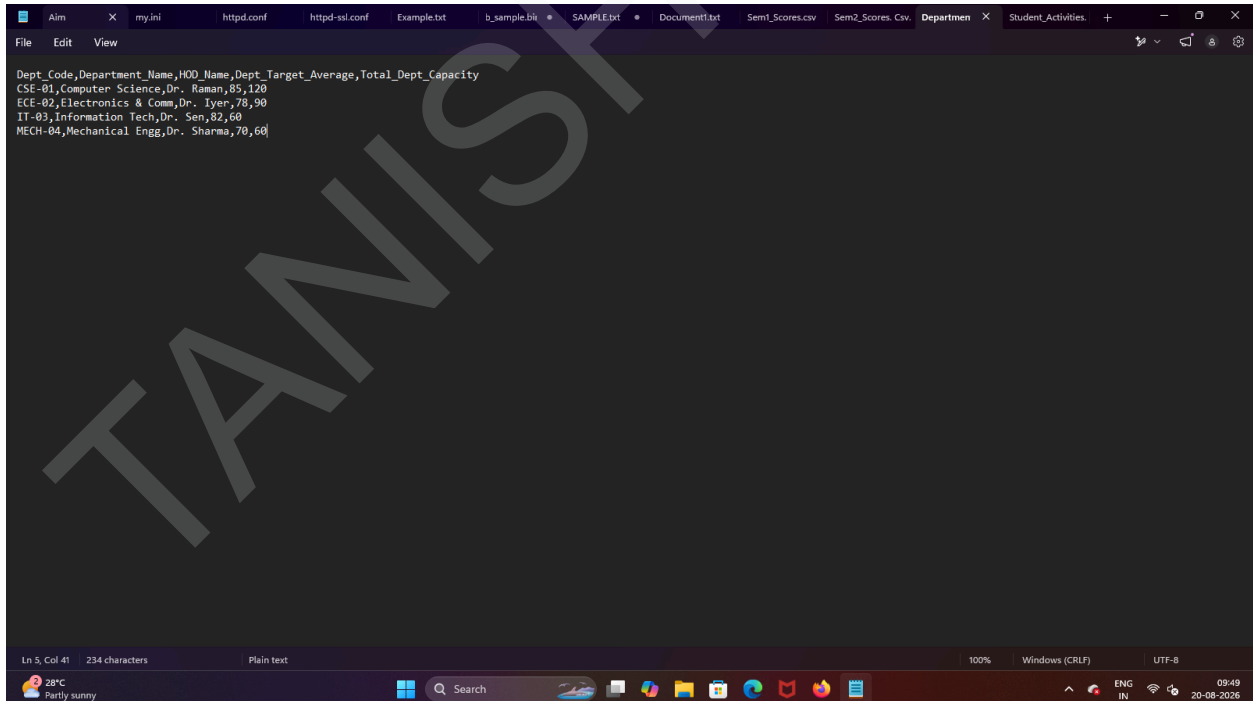
NAME: TANISHKA KUTWAL
ROLL NO: T089

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```
Student_ID,Full_Name,Dept_Info,Math,Physics,Chemistry,Programming,Attendance
S101,Aarav Sharma,CSE-01,88,92%,90,98,88 %
S102,Priya Verma,ECE-02,79,76%,82,88,92 %
S103,Rohan Gupta,CSE-01,82,85%,80,90,80 %
S104,Sneha Patel,IT-03,95,90%,94,98,96 %
S105,Aditya Rao,MECH-04,52,58%,55,65,70 %
S106,Ananya Singh,CSE-01,90,94%,88,95,90 %
S107,Rahul Nair,ECE-02,74,73%,78,82,75 %
S108,Kavya Reddy,IT-03,96,96%,95,99,99 %
S109,Vikram Joshi,CSE-01,70,65%,62,75,82 %
S110,Pooja Iyer,ECE-02,85,88%,84,92,94 %
```

3) Department_Master.csv



```
Dept_Code,Department_Name,HOD_Name,Dept_Target_Average,Total_Dept_Capacity
CSE-01,Computer Science,Dr. Raman,85,120
ECE-02,Electronics & Comm,Dr. Iyer,78,90
IT-03,Information Tech,Dr. Sen,82,60
MECH-04,Mechanical Engg,Dr. Sharma,70,60
```

4) Student_Activities.csv

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```
Student_ID,Club_Membership,Projects_Completed,Certifications
S101,Coding Club,3,2
S102,Robotics Club,2,1
S103,Coding Club,2,1
S104,Debate Society,4,3
S105,Sports Club,1,0
S106,Coding Club,3,2
S107,Robotics Club,2,1
S108,Debate Society,4,3
S109,Sports Club,1,1
S110,Cultural Club,2,2
```

A: Identify and Handle Missing or Inconsistent Data

Step 1: Load the Dataset

1. Open Tableau Desktop.
2. Under the Connect pane on the left, click Text file.
3. Select and open Sem1_Scores.csv.
4. In the Data Source preview table: o Observe missing entries (Null) in Math, Physics, and Chemistry.

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Tableau - Book1

File Data Server Window Help

Connections

Sem1_Scores
Text file

Files

Use Data Interpreter
Data Interpreter might be able to clean your Text file workbook.

Department_Master.csv
Sem1_Scores.csv
Sem2_Scores.Csv.txt
Student_Activities.csv
New Union
New Table Extension

Sem1_Scores.csv

8 fields 10 rows

10 rows

Name Sem1_Scores.csv

Description
No description available.

Fields

Type	Field Name	Physical Table	Remote Field
Abc	Student ID	Sem1_Scores.csv	Student_ID
Abc	Full Name	Sem1_Scores.csv	Full_Name
Abc	Dept Info	Sem1_Scores.csv	Dept_Info
#	Math	Sem1_Scores.csv	Math

Abc	Abc	Abc	#	#	#	#	Abc
Sem1_Scores.csv	Sem1_Scores.csv	Sem1_Scores.csv	Sem1_Scores.csv	Sem1_Scores.csv	Sem1_Scores.csv	Sem1_Scores.csv	Sem1_Scores.csv
Student ID	Full Name	Dept Info	Math	Physics	Chemistry	Programming	Attendance
S101	Aarav Sharma	CSE-01	85	0.900000	88	95	85 %
S102	priya VERMA	ECE-02	null	0.750000	80	85	90 %
S103	Rohan Gupta	CSE-01	78	0.820000	null	88	78 %
S104	Sneha Patel	IT-03	92	0.880000	90	96	95 %
S105	aditya Rao	MECH-04	45	null	50	60	65 %
S106	Ananya Singh	CSE-01	88	0.910000	85	92	88 %
S107	Rahul Nair	ECE-02	null	0.700000	75	80	72 %

Go to Worksheet

Data Source Sheet 1

28°C Partly sunny 09:50 20-08-2026

Step 2: Clean Inconsistent String Fields (Convert Text Percentages to Numeric Measures)

1. Navigate to Sheet 1 (bottom-left tab).

2. Right-click an empty space in the Data Pane (left sidebar) --> select Create Calculated Field....

3. Create the cleaned measure for Physics:

Field Name: Physics_Clean o Tableau Formula: Code snippet

FLOAT(REPLACE(STR([Physics]), '%', ''))

Click OK.

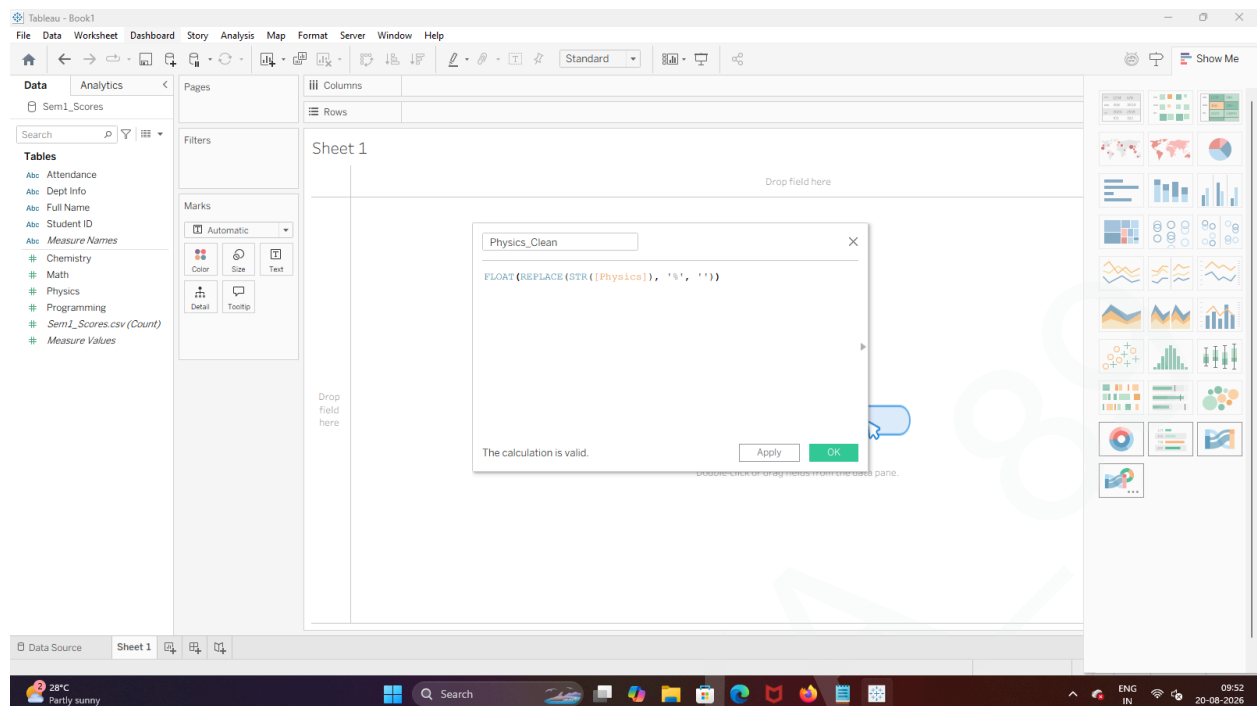
NAME: TANISHKA KUTWAL

ROLL NO: T089

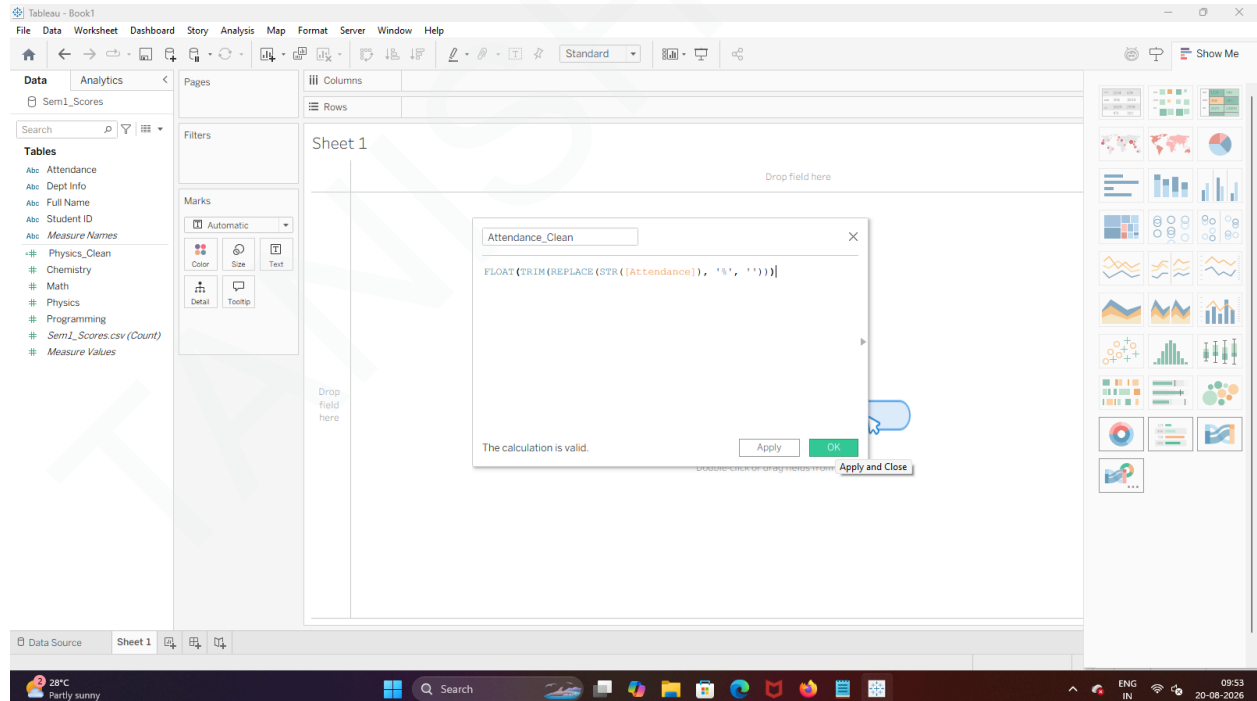
SHETH L.U.J AND SIR M.V COLLEGE

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4. Create the cleaned measure for Attendance: o Right-click the Data Pane --> Create Calculated Field.... o Field Name: Attendance_Clean o Tableau Formula: Code snippet `FLOAT(TRIM(REPLACE(STR([Attendance]), '%', '')))` o Click OK.



Step 3: Handle Missing (Null) Numeric Values via Imputatio

1. Handle missing entries in Math: o Right-click the Data Pane --> Create Calculated Field.... o Field Name: Math_Clean o Tableau Formula: Code snippet `IFNULL([Math], 0)` o Click OK.

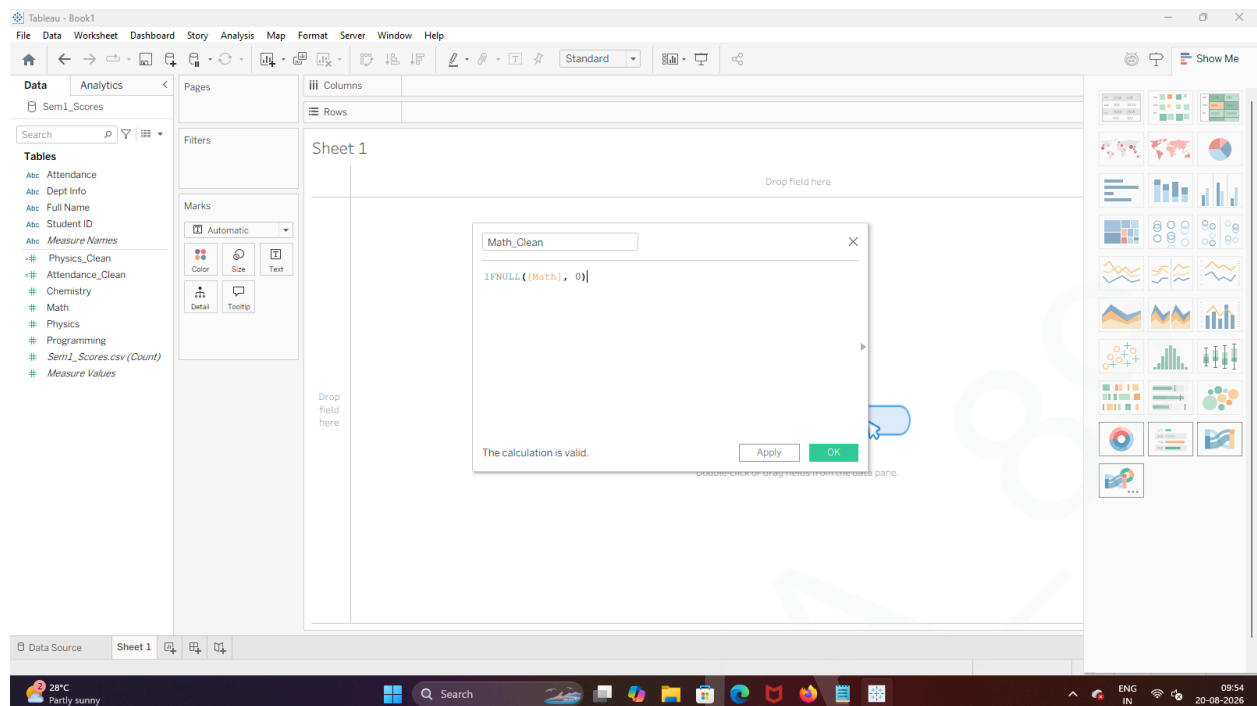
NAME: TANISHKA KUTWAL

ROLL NO: T089

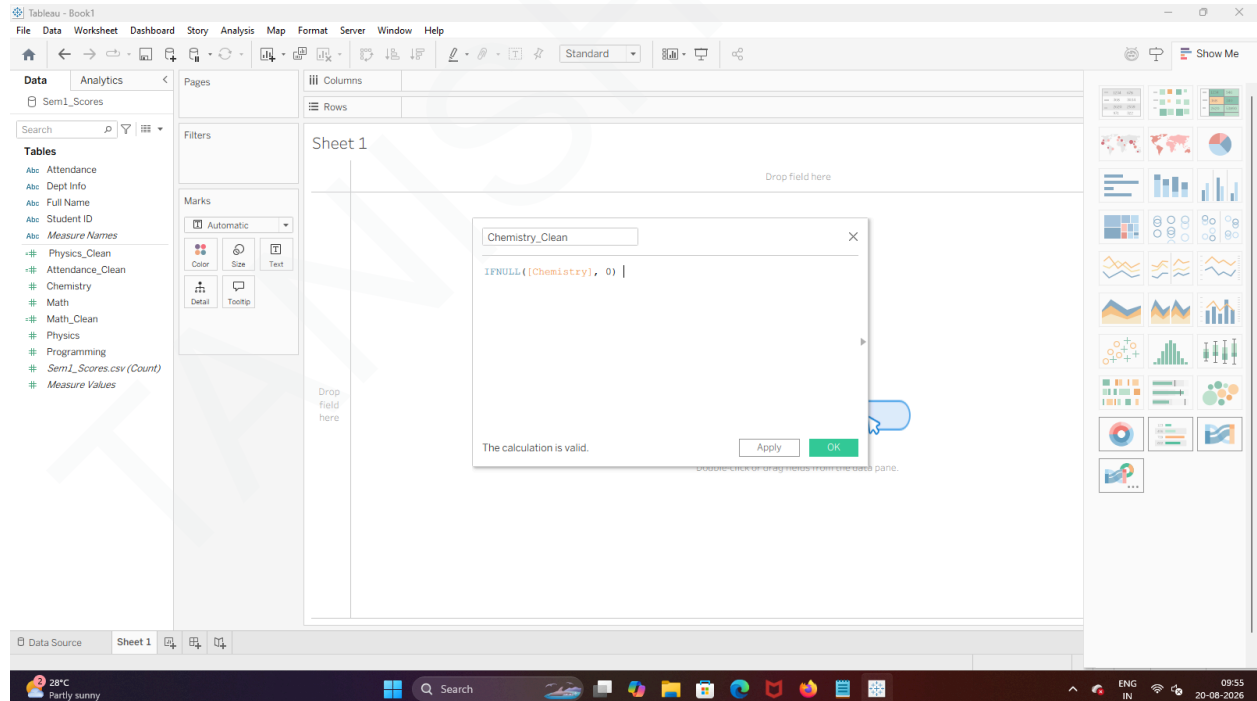
SHETH L.U.J AND SIR M.V COLLEGE

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2. Handle missing entries in Chemistry: o Right-click the Data Pane --> Create Calculated Field.... o Field Name: Chemistry_Clean o Tableau Formula: Code snippet IFNULL([Chemistry], 0) o Click OK.



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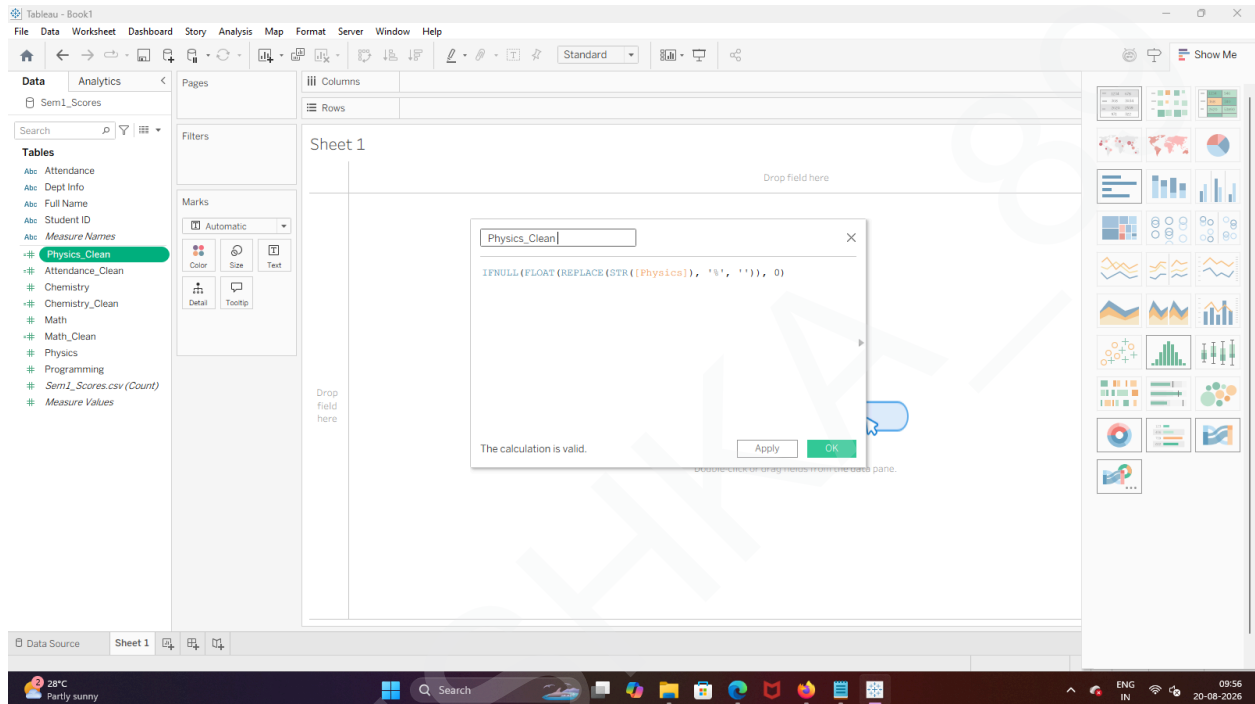
Step 4: Impute Missing Values in Physics (Optional Handling for Student S105)

1. Student S105 contains a Null value in Physics. Combine string stripping with null handling:

o Edit Physics_Clean (right-click Physics_Clean --> Edit...):

o Updated Tableau Formula: Code snippet IFNULL(FLOAT(REPLACE(STR([Physics]), '%', '')), 0)

oClick OK.



B: Convert Wide Data into Tall Format

Step 1: Open the Data Source Grid 1. In Tableau, navigate to the Data Source tab at the bottom-left corner

. 2. Locate the four horizontal subject score columns: o Math o Physics o Chemistry o Programming

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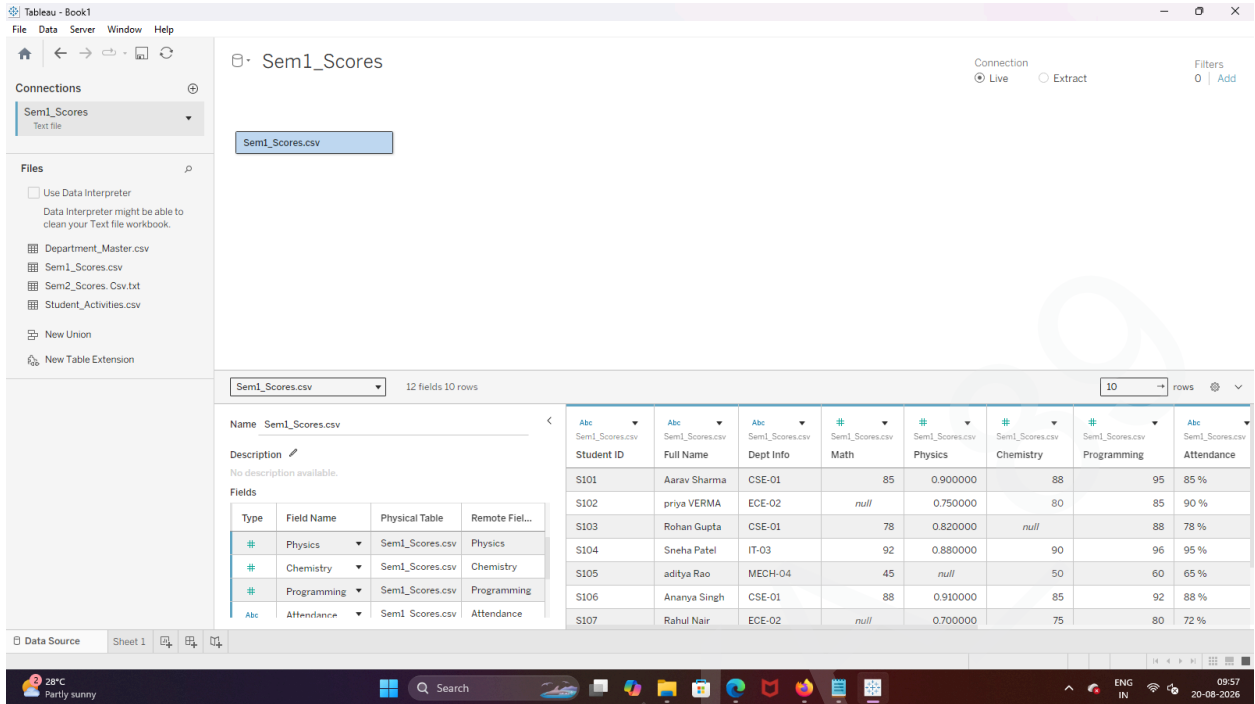


Tableau - Book1

File Data Server Window Help

Connections

Sem1_Scores

Text file

Files

Use Data Interpreter

Data Interpreter might be able to clean your Text file workbook.

Department_Master.csv

Sem1_Scores.csv

Sem2_Scores.Csv.txt

Student_Activities.csv

New Union

New Table Extension

Sem1_Scores

Connection

Live Extract

Filters

0 Add

Sem1_Scores.csv

12 fields 10 rows

10 rows

Name Sem1_Scores.csv

Description

No description available.

Fields

Type	Field Name	Physical Table	Remote Fiel...
#	Physics	Sem1_Scores.csv	Physics
#	Chemistry	Sem1_Scores.csv	Chemistry
#	Programming	Sem1_Scores.csv	Programming
Abc	Attendance	Sem1_Scores.csv	Attendance

Abc	Abc	Abc	#	#	#	#	Abc
Sem1_Scores.csv	Sem1_Scores.csv	Sem1_Scores.csv	Sem1_Scores.csv	Sem1_Scores.csv	Sem1_Scores.csv	Sem1_Scores.csv	Sem1_Scores.csv
Student ID	Full Name	Dept Info	Math	Physics	Chemistry	Programming	Attendance
S101	Aarav Sharma	CSE-01	85	0.900000	88	95	85 %
S102	priya VERMA	ECE-02	null	0.750000	80	85	90 %
S103	Rohan Gupta	CSE-01	78	0.820000	null	88	78 %
S104	Sneha Patel	IT-03	92	0.880000	90	96	95 %
S105	aditya Rao	MECH-04	45	null	50	60	65 %
S106	Ananya Singh	CSE-01	88	0.910000	85	92	88 %
S107	Rahul Nair	ECE-02	null	0.700000	75	80	72 %

Data Source

Sheet 1

28°C Partly sunny

Search

ENG IN

09:57 20-08-2026

Step 2: Multi-Select and Pivot Columns

1. Click the header of the Math column.
2. Hold down Ctrl (or Cmd on Mac) and click Physics, Chemistry, and Programming so all four columns are highlighted in blue.

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Tableau - Book1

File Data Server Window Help

Connections

Sem1_Scores

Text file

Files

Use Data Interpreter

Data Interpreter might be able to clean your Text file workbook.

Department_Master.csv

Sem1_Scores.csv

Sem2_Scores.Csv.txt

Student_Activities.csv

New Union

New Table Extension

Sem1_Scores

12 fields 10 rows

10 rows

Name Sem1_Scores.csv

Description

No description available.

Fields

Type	Field Name	Physical Table	Remote Field
Alt	Dept Info	Sem1_Scores.csv	Dept Info
Alt	Math	Sem1_Scores.csv	Math
Alt	Physics	Sem1_Scores.csv	Physics
Alt	Chemistry	Sem1_Scores.csv	Chemistry
Alt	Programming	Sem1_Scores.csv	Programming
Alt	Attendance	Sem1_Scores.csv	Attendance

Student ID	Full Name	Dept Info	Math	Physics	Chemistry	Programming	Attendance	Physics_Clean	Attendance_Clean
S101	Aarav Sharma	CSE-01	85	0.900000	88	95	85%	0.900000	85.0000
S102	priya VERMA	ECE-02	null	0.750000	80	85	90%	0.750000	90.0000
S103	Rohan Gupta	CSE-01	78	0.820000	null	88	78%	0.820000	78.0000
S104	Sneha Patel	IT-03	92	0.880000	90	96	95%	0.880000	95.0000
S105	aditya Rao	MECH-04	45	null	50	60	65%	0.000000	65.0000
S106	Ananya Singh	CSE-01	88	0.910000	85	92	88%	0.910000	88.0000
S107	Rahul Nair	ECE-02	null	0.700000	75	80	72%	0.700000	72.0000
S108	Kavya Reddy	IT-03	95	0.940000	92	98	98%	0.940000	98.0000
S109	Vikram Joshi	CSE-01	67	0.600000	58	72	80%	0.600000	80.0000
S110	Devika Joshi	ECE-02	83	0.850000	80	80	81%	0.850000	81.0000

3. Right-click on any of the highlighted column headers. 4. Select Pivot.

Tableau - Book1

File Data Server Window Help

Connections

Sem1_Scores

Text file

Files

Use Data Interpreter

Data Interpreter might be able to clean your Text file workbook.

Department_Master.csv

Sem1_Scores.csv

Sem2_Scores.Csv.txt

Student_Activities.csv

New Union

New Table Extension

Sem1_Scores

12 fields 10 rows

10 rows

Name Sem1_Scores.csv

Description

No description available.

Fields

Type	Field Name	Physical Table	Remote Field
Alt	Dept Info	Sem1_Scores.csv	Dept Info
Alt	Math	Sem1_Scores.csv	Math
Alt	Physics	Sem1_Scores.csv	Physics
Alt	Chemistry	Sem1_Scores.csv	Chemistry
Alt	Programming	Sem1_Scores.csv	Programming
Alt	Attendance	Sem1_Scores.csv	Attendance

Student ID	Full Name	Dept Info	Math	Physics	Chemistry	Programming	Attendance	Physics_Clean	Attendance_Clean
S101	Aarav Sharma	CSE-01	85	0.900000	88	95	85%	0.900000	85.0000
S102	priya VERMA	ECE-02	null	0.750000	80	85	90%	0.750000	90.0000
S103	Rohan Gupta	CSE-01	78	0.820000	null	88	78%	0.820000	78.0000
S104	Sneha Patel	IT-03	92	0.880000	90	96	95%	0.880000	95.0000
S105	aditya Rao	MECH-04	45	null	50	60	65%	0.000000	65.0000
S106	Ananya Singh	CSE-01	88	0.910000	85	92	88%	0.910000	88.0000
S107	Rahul Nair	ECE-02	null	0.700000	75	80	72%	0.700000	72.0000
S108	Kavya Reddy	IT-03	95	0.940000	92	98	98%	0.940000	98.0000
S109	Vikram Joshi	CSE-01	67	0.600000	58	72	80%	0.600000	80.0000
S110	Devika Joshi	ECE-02	83	0.850000	80	80	81%	0.850000	81.0000

Step 3: Rename the Pivoted Fields Tableau replaces the four columns with two consolidated columns at the end of the table:

1. Double-click the header named Pivot Field Names and type: Subject
2. Double-click the header named Pivot Field Values and type: Score

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Tableau - Book1

File Data Server Window Help

Connections

Sem1_Scores
Text file

Files

Use Data Interpreter
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Department_Master.csv
Sem1_Scores.csv
Sem2_Scores.Csv.txt
Student_Activities.csv
New Union
New Table Extension

Sem1_Scores

Connection
Live Extract
Filters
0 Add

Relate tables
Drag tables near Sem1_Scores.csv to relate them

Add a base table
Drag out additional base tables for multi-fact analysis.

Need help adding data? Learn more

Sem1_Scores.csv 10 fields 40 rows

Name Sem1_Scores.csv
Description No description available.

Fields

Type	Field Name	Physical Table	Remote Field
Abc	Attendance	Sem1_Scores.csv	Attendance
Abc	Subject	Pivot	Pivot Field Na...
+	Score	Pivot	Pivot Field Val...
+	Physics_Clean	Calculation	Calculation_1...
+	Attendance_C	Calculation	Calculation_1...
+	Math_Clean	Calculation	Calculation_1...

Abc	Abc	Abc	Abc	Abc	+ <th>+ <th>+ <th>+ <th>+ </th></th></th></th>	+ <th>+ <th>+ <th>+ </th></th></th>	+ <th>+ <th>+ </th></th>	+ <th>+ </th>	+
Sem1_Scores.csv	Sem1_Scores.csv	Sem1_Scores.csv	Sem1_Scores.csv	Pivot	Pivot	Calculation	Calculation	Calculation	Calculation
Student ID	Full Name	Dept Info	Attendance	Subject	Score	Physics_Clean	Attendance_Clean	Math_Clean	Chem
S101	Aarav Sharma	CSE-01	85 %	Chemistry	88.0000	null	85.0000	null	
S101	Aarav Sharma	CSE-01	85 %	Math	85.0000	null	85.0000	null	
S101	Aarav Sharma	CSE-01	85 %	Physics	0.9000	null	85.0000	null	
S101	Aarav Sharma	CSE-01	85 %	Programming	95.0000	null	85.0000	null	
S102	priya VERMA	ECE-02	90 %	Chemistry	80.0000	null	90.0000	null	
S102	priya VERMA	ECE-02	90 %	Math	null	null	90.0000	null	
S102	priya VERMA	ECE-02	90 %	Physics	0.7500	null	90.0000	null	
S102	priya VERMA	ECE-02	90 %	Programming	85.0000	null	90.0000	null	
S103	Rohan Gupta	CSE-01	78 %	Chemistry	null	null	78.0000	null	
S103	Rohan Gupta	CSE-01	78 %	Math	78.0000	null	78.0000	null	
S103	Rohan Gupta	CSE-01	78 %	Physics	0.8200	null	78.0000	null	
S103	Rohan Gupta	CSE-01	78 %	Programming	88.0000	null	78.0000	null	

Data Source Sheet 1

Step 4: Clean and Format the Tall Score Field Because raw Physics contained % symbols, the new combined Score column defaults to text (Abc). Convert it to a numeric measure:

1. Click the Abc icon directly above the Score column header.
2. Select Number (decimal) from the dropdown list.

Tableau - Book1

File Data Server Window Help

Connections

Sem1_Scores
Text file

Files

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Department_Master.csv
Sem1_Scores.csv
Sem2_Scores.Csv.txt
Student_Activities.csv
New Union
New Table Extension

Sem1_Scores

Connection
Live Extract
Filters
0 Add

Sem1_Scores.csv 10 fields 40 rows

Name Sem1_Scores.csv
Description No description available.

Fields

Type	Field Name	Physical Table	Remote Field
Number (decimal)	Score	Pivot	Pivot Field Val...
+	Physics_Clean	Calculation	Calculation_1...
+	Attendance_C	Calculation	Calculation_1...
+	Math_Clean	Calculation	Calculation_1...

Abc	Abc	Abc	Abc	Abc	+ <th>+ <th>+ <th>+ <th>+ </th></th></th></th>	+ <th>+ <th>+ <th>+ </th></th></th>	+ <th>+ <th>+ </th></th>	+ <th>+ </th>	+
Sem1_Scores.csv	Sem1_Scores.csv	Sem1_Scores.csv	Sem1_Scores.csv	Pivot	Pivot	Calculation	Calculation	Calculation	Calculation
Student ID	Full Name	Dept Info	Attendance	Subject	Score	Physics_Clean	Attendance_Clean	Math_Clean	Chem
S101	Aarav Sharma	CSE-01	85 %	Chemistry	88.0000	88.0000	85.0000	null	
S101	Aarav Sharma	CSE-01	85 %	Math	85.0000	85.0000	85.0000	null	
S101	Aarav Sharma	CSE-01	85 %	Physics	0.9000	0.9000	85.0000	null	
S101	Aarav Sharma	CSE-01	85 %	Programming	95.0000	95.0000	85.0000	null	
S102	priya VERMA	ECE-02	90 %	Chemistry	80.0000	80.0000	90.0000	null	
S102	priya VERMA	ECE-02	90 %	Math	null	null	90.0000	null	
S102	priya VERMA	ECE-02	90 %	Physics	0.7500	0.7500	90.0000	null	
S102	priya VERMA	ECE-02	90 %	Programming	85.0000	85.0000	90.0000	null	
S103	Rohan Gupta	CSE-01	78 %	Chemistry	null	null	78.0000	null	
S103	Rohan Gupta	CSE-01	78 %	Math	78.0000	78.0000	78.0000	null	
S103	Rohan Gupta	CSE-01	78 %	Physics	0.8200	0.8200	78.0000	null	
S103	Rohan Gupta	CSE-01	78 %	Programming	88.0000	88.0000	78.0000	null	

Data Source Sheet 1

3. Alternatively, create a calculated field inside Sheet

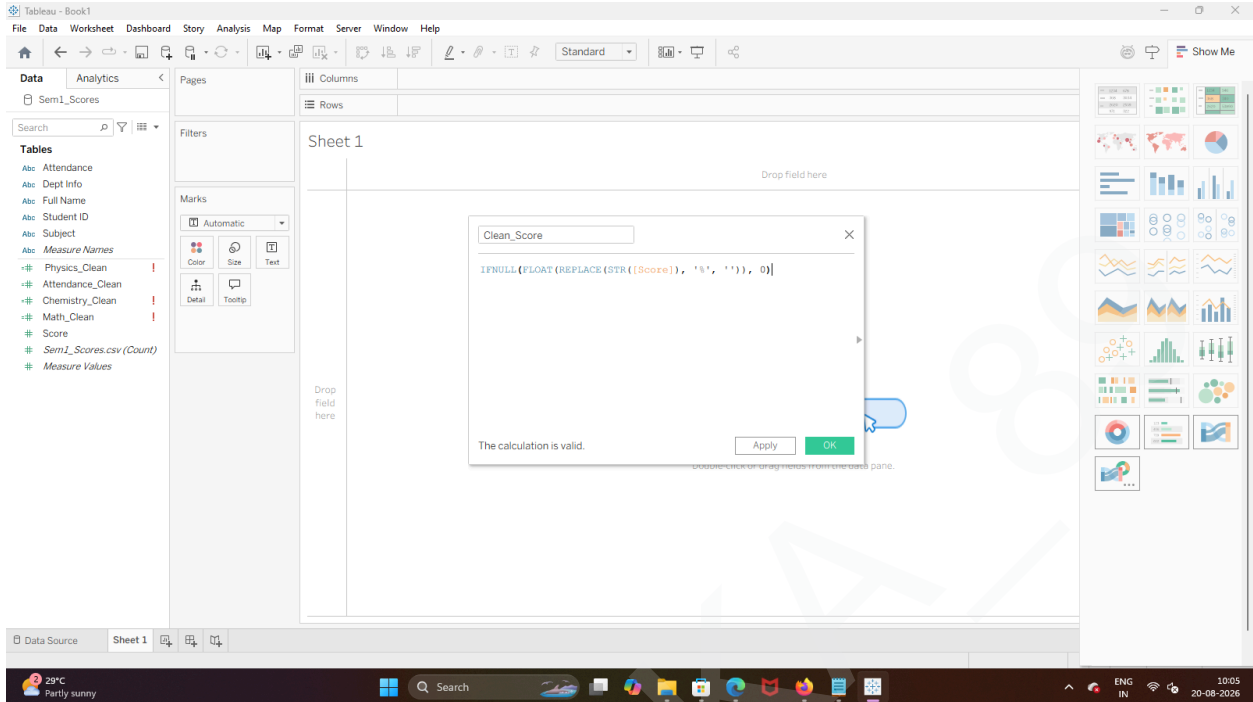
1 named Clean_Score:

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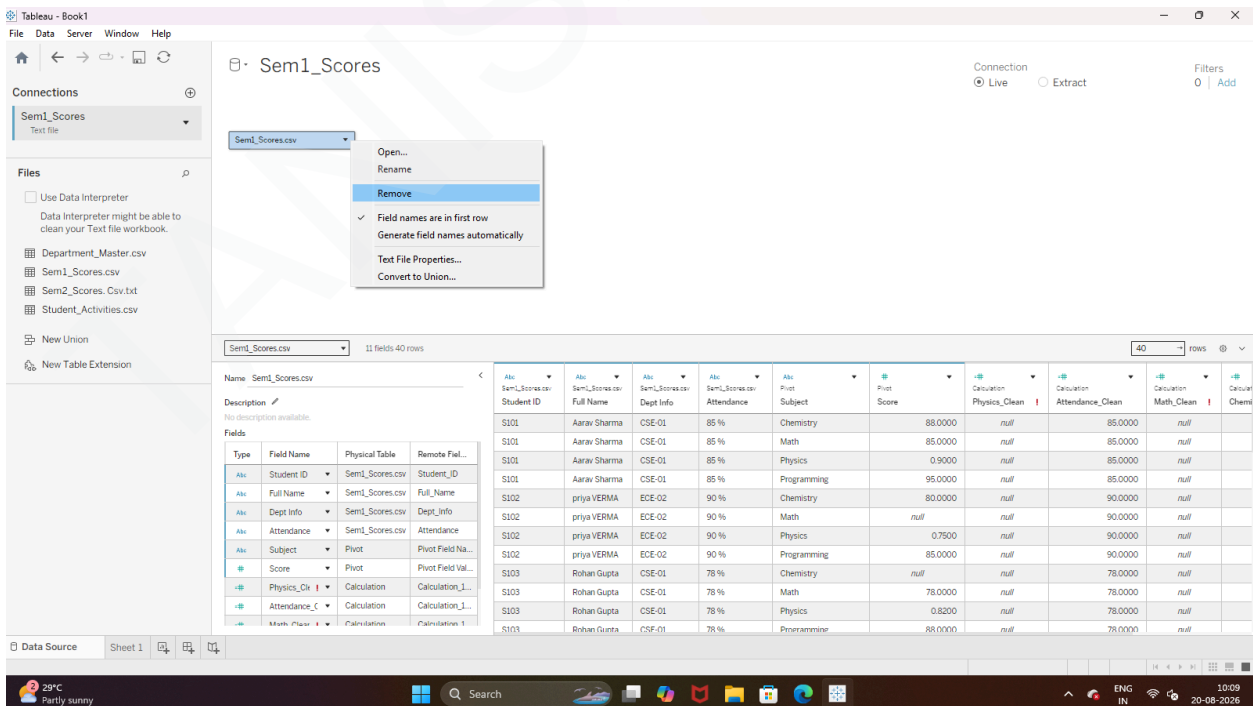
Code snippet IFNULL(FLOAT(REPLACE(STR([Score]), '%', '')), 0)



C: Perform Union of Multiple Files

Step 1: Open the Data Source Canvas

1. Open Tableau Desktop and go to the Data Source tab in the bottom-left corner.
2. If any single table is already on the canvas, right-click it and select Remove to start with a clean canvas.

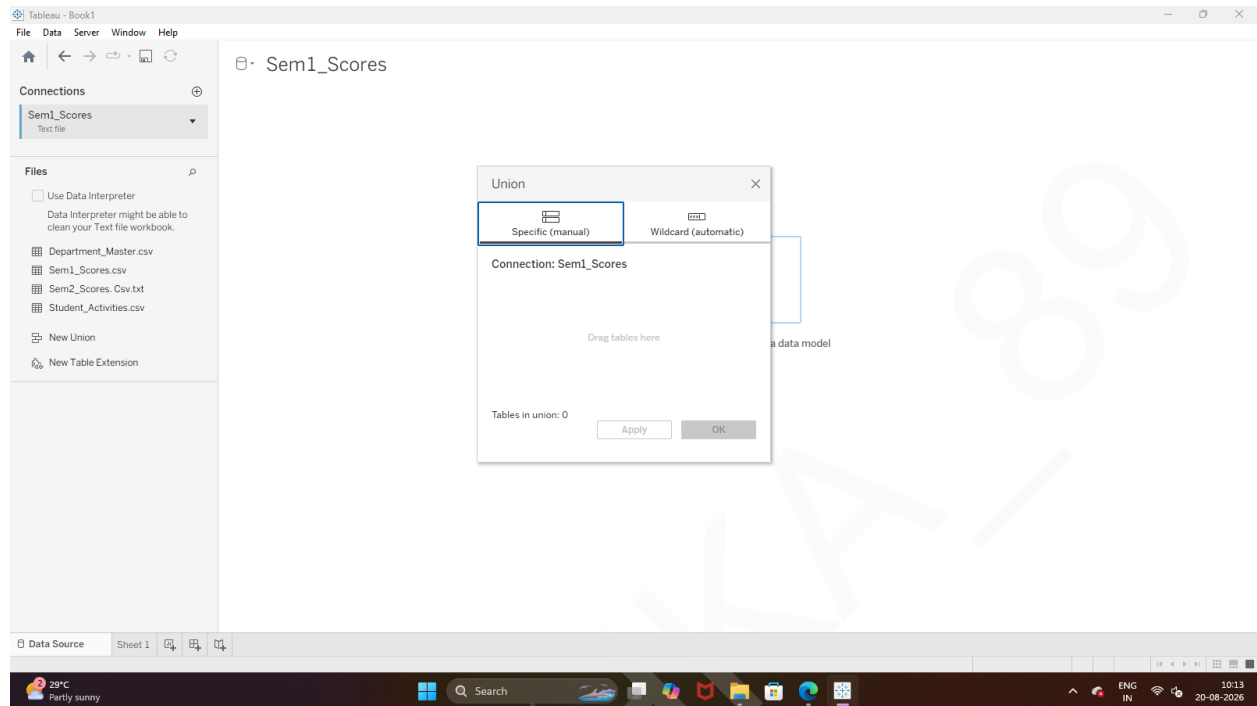


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ROLL NO: T089

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Step 2: Create a Multi-File Union

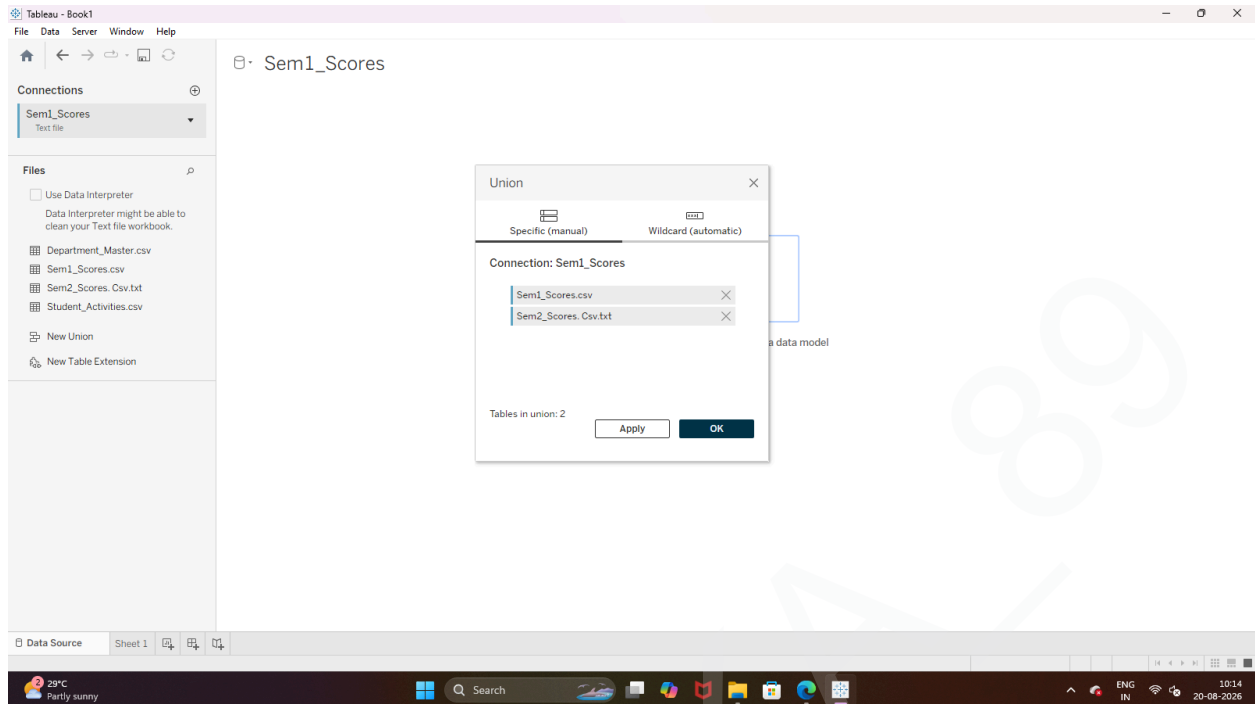
1. Look at the left sidebar under the Files / Connections pane.
2. Drag the New Union option onto the empty canvas area.



3. In the pop-up Union dialog box:
 - o Drag Sem1_Scores.csv from the left panel into the box.
 - o Drag Sem2_Scores.csv from the left panel into the same box.
 - o Verify that both filenames are listed in the Union box.
4. Click Apply, then click OK.

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Step 3: Verify the Merged Data Grid

1. Review the data grid at the bottom of the screen.
2. Confirm the row count: the total number of rows increases from 10 to 20 rows (10 records for Semester 1 and 10 records for Semester 2).
3. Scroll to the far right of the grid to verify the generated system column:
 - o Table Name: Displays "Sem1_Scores.csv" for the first 10 rows and "Sem2_Scores.csv" for the remaining 10 rows.

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The screenshot shows the Tableau Desktop interface. The 'Connections' pane on the left lists 'Sem1_Scores' as a text file. The 'Files' pane lists several CSV and TXT files. The central workspace shows a 'Union' icon. The data preview table at the bottom displays the following data:

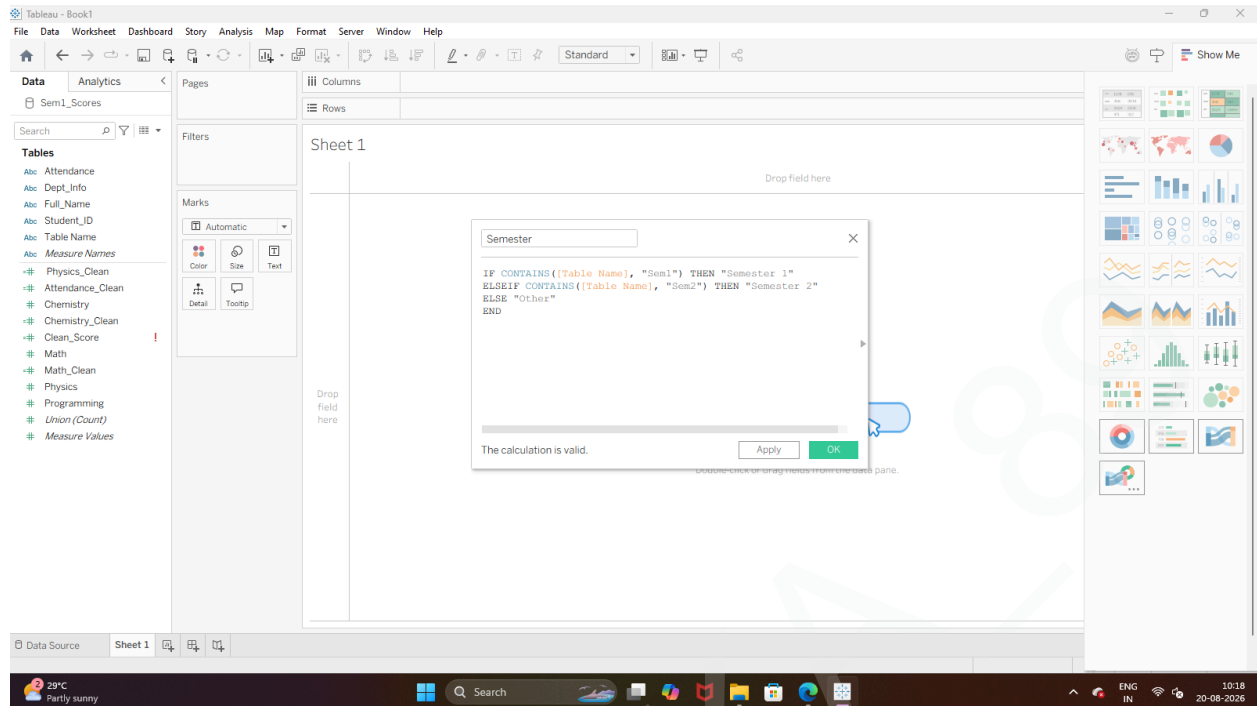
Student_ID	Full_Name	Dept_Info	Math	Physics	Chemistry	Programming	Attendance	Table Name	Physics_Clean	Attendance
S101	Aarav Sharma	CSE-01	85	0.900000	88	95	85 %	Sem1_Scores.csv	0.900000	
S102	priya VERMA	ECE-02	null	0.750000	80	85	90 %	Sem1_Scores.csv	0.750000	
S103	Rohan Gupta	CSE-01	78	0.820000	null	88	78 %	Sem1_Scores.csv	0.820000	
S104	Sneha Patel	IT-03	92	0.880000	90	96	95 %	Sem1_Scores.csv	0.880000	
S105	aditya Rao	MECH-04	45	null	50	60	65 %	Sem1_Scores.csv	0.000000	
S106	Ananya Singh	CSE-02	88	0.910000	85	92	88 %	Sem1_Scores.csv	0.910000	
S107	Rahul Nair	ECE-02	null	0.700000	75	80	72 %	Sem1_Scores.csv	0.700000	
S108	Kavya Reddy	IT-03	95	0.940000	92	98	98 %	Sem1_Scores.csv	0.940000	
S109	Vikram Joshi	CSE-01	67	0.600000	58	72	80 %	Sem1_Scores.csv	0.600000	

Step 4: Create a Clean Semester Identifier Dimension

1. Click on Sheet 1 at the bottom-left.
2. In the Data Pane (left sidebar), right-click in an empty space select Create Calculated Field....
3. Set the Field Name to: Semester
4. Enter the following formula:
IF CONTAINS([Table Name], "Sem1") THEN "Semester 1"
ELSEIF CONTAINS([Table Name], "Sem2") THEN "Semester 2"
ELSE "Other"
END

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D: Execute Cross-Database Joins

Step 1:

Stay on the Top Canvas Leave your Union table box as it is on the canvas.

Step 2:

Add and Link Department_Master.csv 1. In the left sidebar under Files / Connections, find Department_Master.csv.

2. Drag Department_Master.csv onto the canvas and drop it to the right of your Union box.

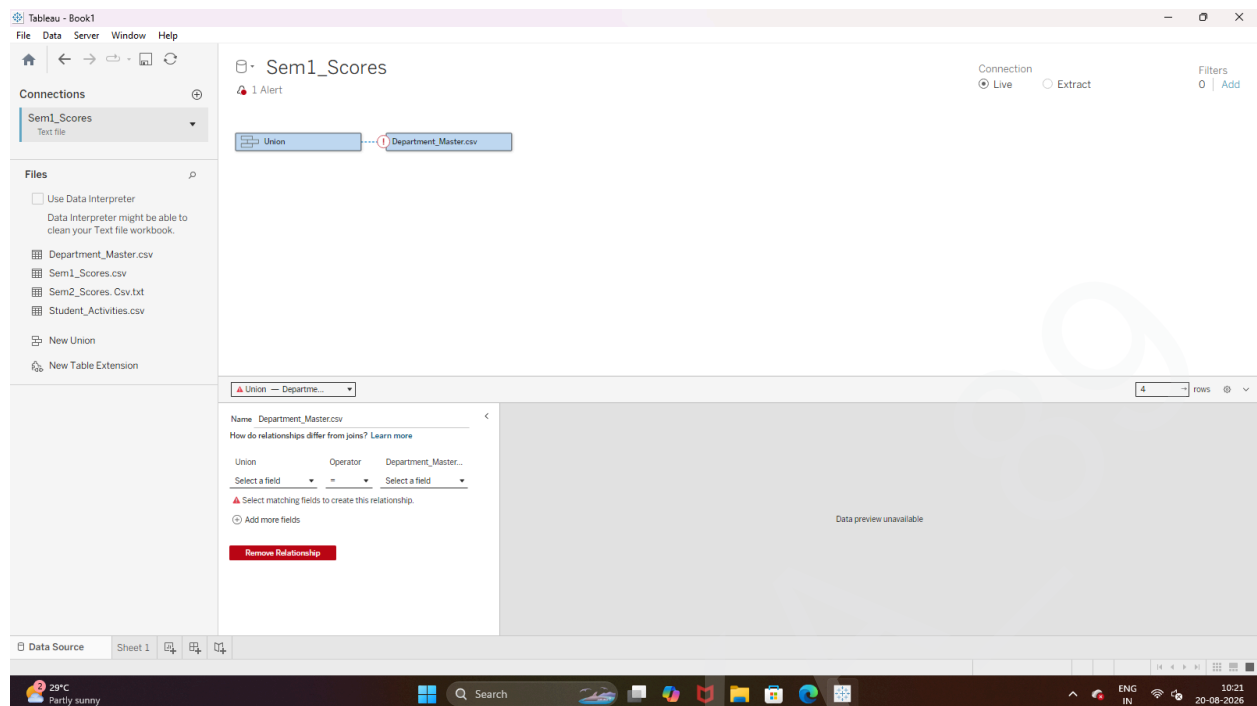
3. A flexible line (relationship / join noodle) will connect the two boxes.

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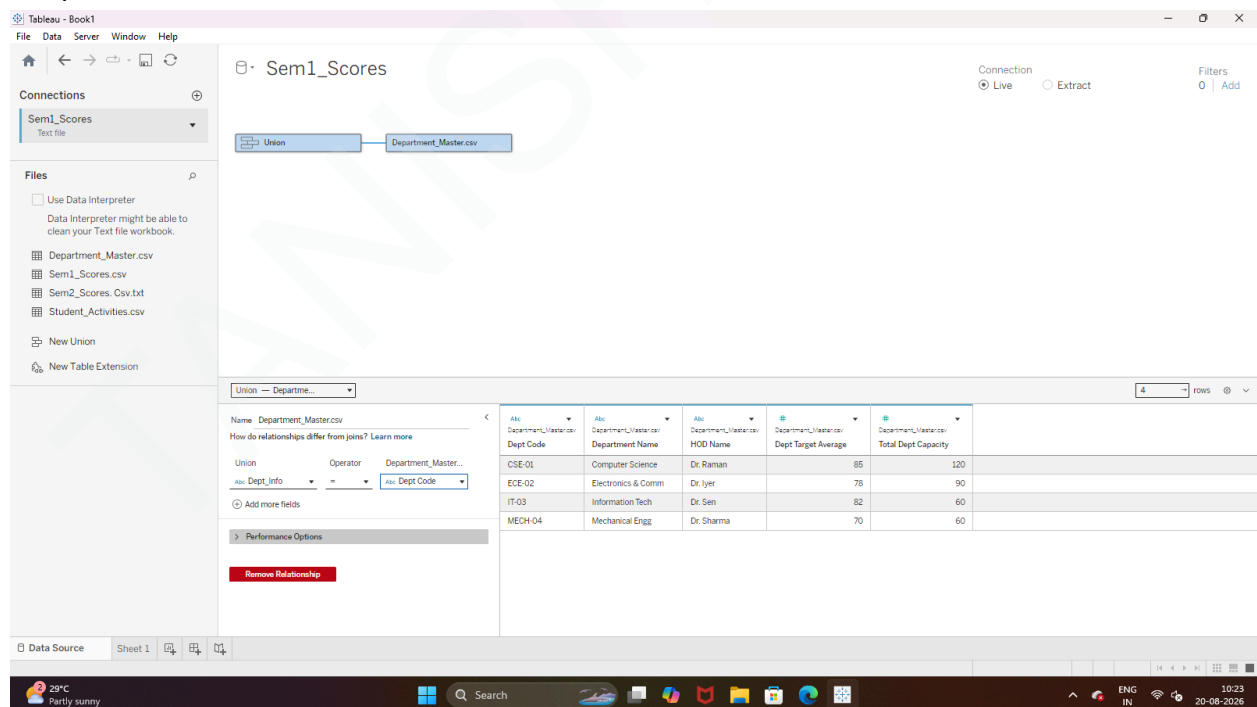
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4. If the relationship dialog opens (or click the relationship line):
- o Left Table Field (Union): Select Dept_Info
 - o Right Table Field (Department_Master): Select Dept_Code
 - o Operator: =



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Step 3: Add and Link Student_Activities.csv

1. From the left sidebar, drag Student_Activities.csv onto the canvas.
2. Drop it to the right of the Union box.

The screenshot shows the Tableau Desktop interface. In the left sidebar, the 'Connections' pane shows 'Sem1_Scores' as a text file. The 'Files' pane lists several CSV and TXT files, including 'Department_Master.csv', 'Sem1_Scores.csv', 'Sem2_Scores.Csv.txt', and 'Student_Activities.csv'. The main canvas displays a 'Union' relationship between 'Department_Master.csv' and 'Student_Activities.csv'. The 'Relationship' dialog is open, showing the 'Student ID' field selected for both tables with an equals operator. The data preview shows 10 rows of student activity data.

Student ID (Student Activities)	Club Membership	Projects Completed	Certifications
S301	Coding Club	3	2
S302	Robotics Club	2	1
S303	Coding Club	2	1
S304	Debate Society	4	3
S305	Sports Club	1	0
S306	Coding Club	3	2
S307	Robotics Club	2	1
S308	Debate Society	4	3
S309	Sports Club	1	1
S310	Cultural Club	2	2

3. In the relationship dialog:

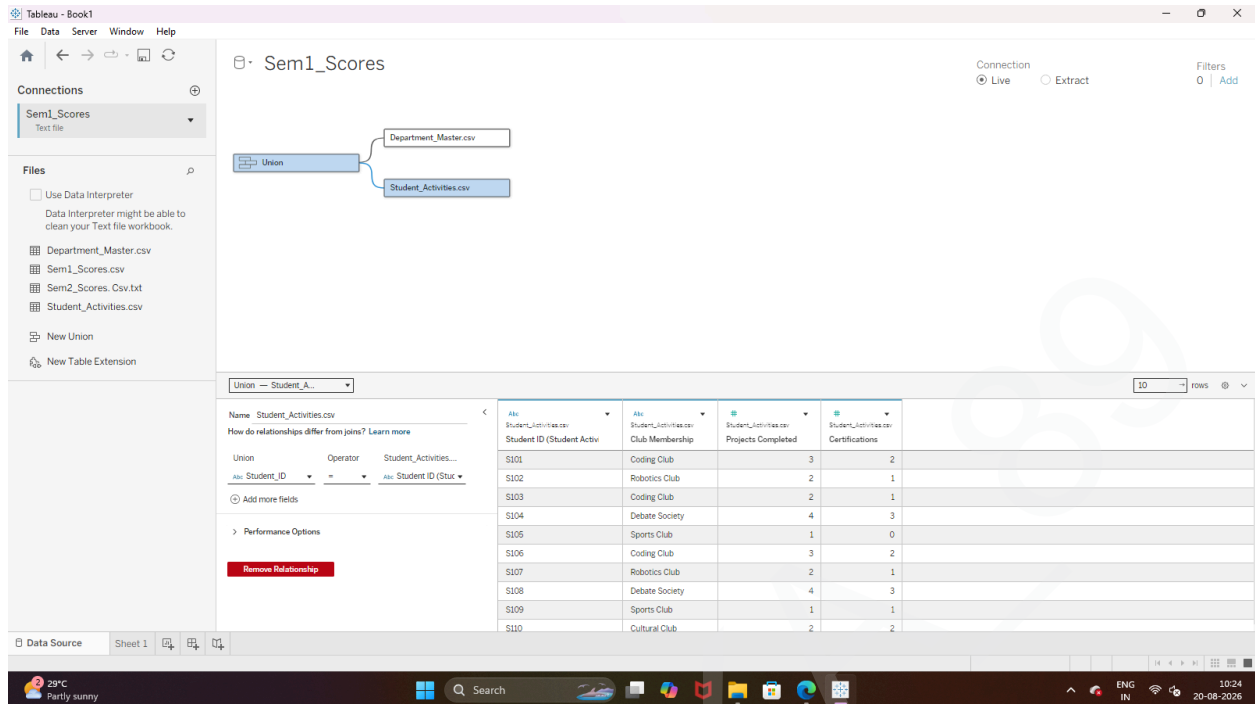
- o Left Table Field (Union): Select Student_ID
- o Right Table Field (Student_Activities): Select Student_ID
- o Operator: =

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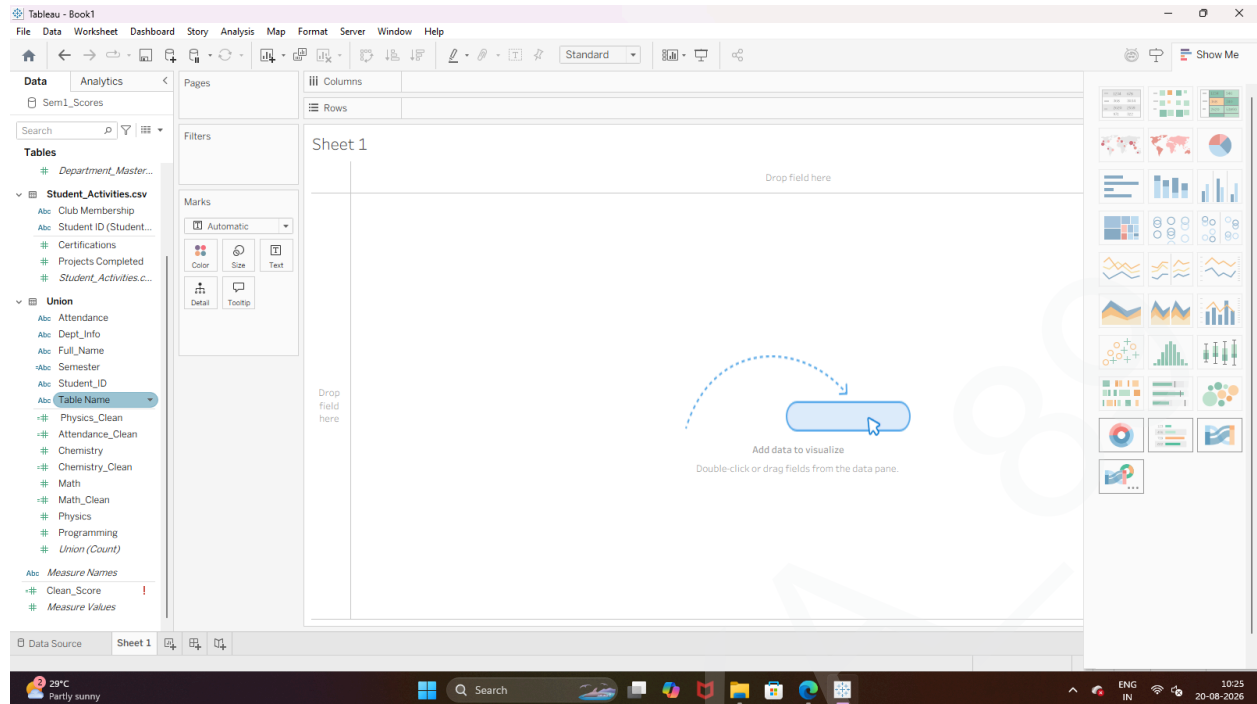


Step 4: Verify in Sheet 1

1. Click on Sheet 1 at the bottom-left.
2. In the Data Pane on the far left, you will now see all three tables organized with their respective fields:
 - o Union: Student_ID, Full_Name, Dept_Info, Math, Physics, Chemistry, Programming, Attendance, Table Name
 - o Department_Master: Dept_Code, Department_Name, HOD_Name, Dept_Target_Average, Total_Dept_Capacity
 - o Student_Activities: Club_Membership, Projects_Completed, Certifications

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E: Restructure Data for Better Visualization

Step 1: Calculate Total Marks and Scaled Percentage

1. Navigate to Sheet 1 in Tableau

2. In the Data Pane on the left, right-click an empty space --> select Create Calculated Field....

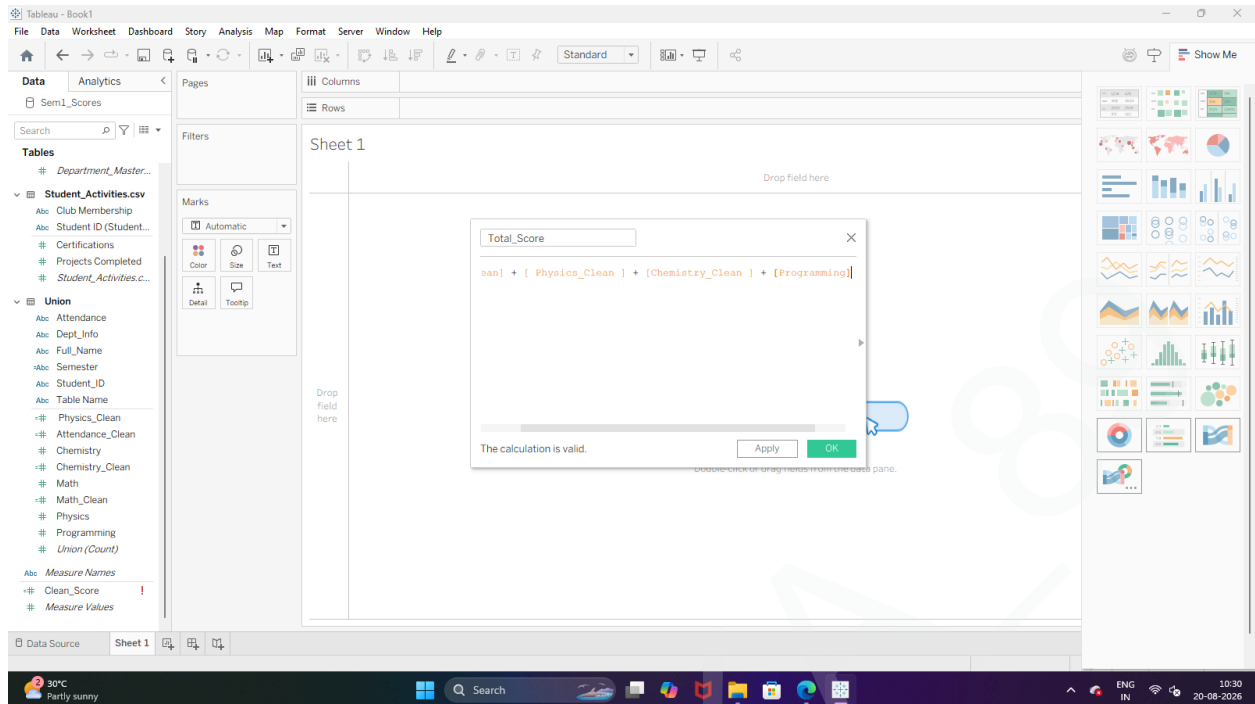
3. Create the student's total marks field: o Field Name: Total_Score o Tableau Formula: Code snippet `[Math_Clean] + [Physics_Clean] + [Chemistry_Clean] + [Programming]`

o Click OK.

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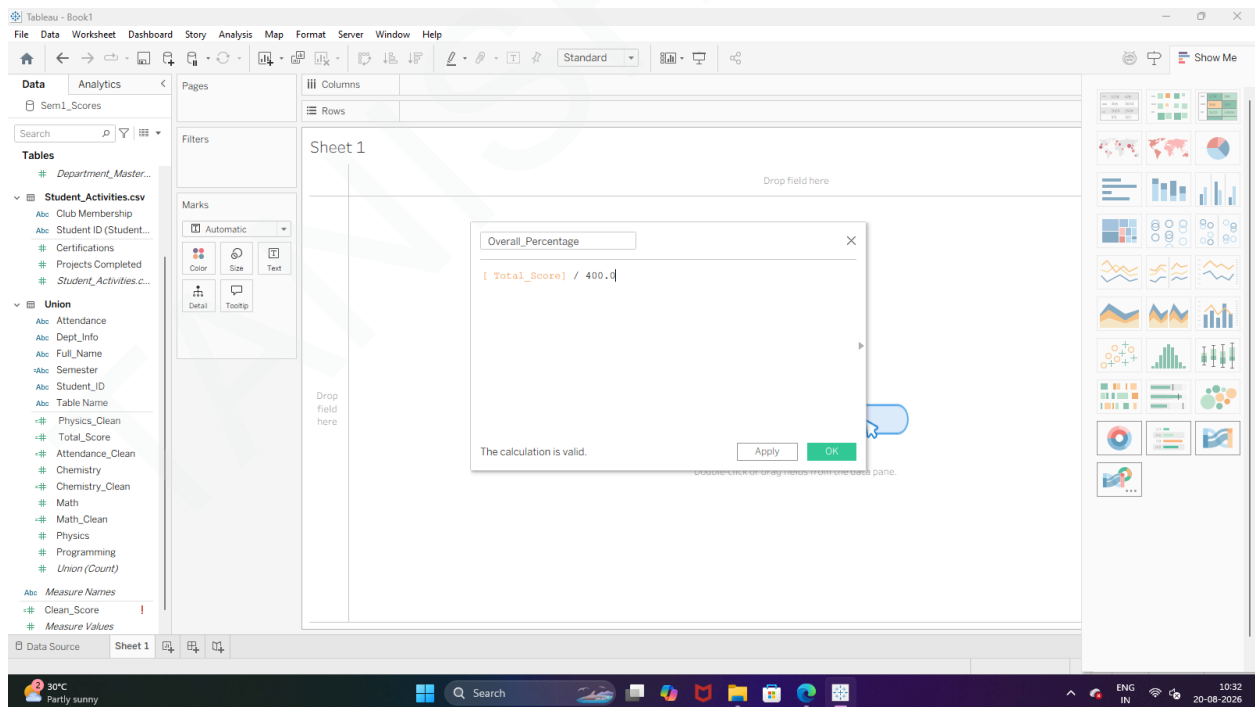
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4. Create the scaled overall percentage field:

- o Right-click in the Data Pane --> select Create Calculated Field....

- o Field Name: Overall_Percentage o Tableau Formula: Code snippet $\text{[Total_Score]} / 400.0$ o Click OK.



5. Set the default percentage display formatting:

- o In the Data Pane, right-click Overall_Percentage -->Default Properties - >Number Format....

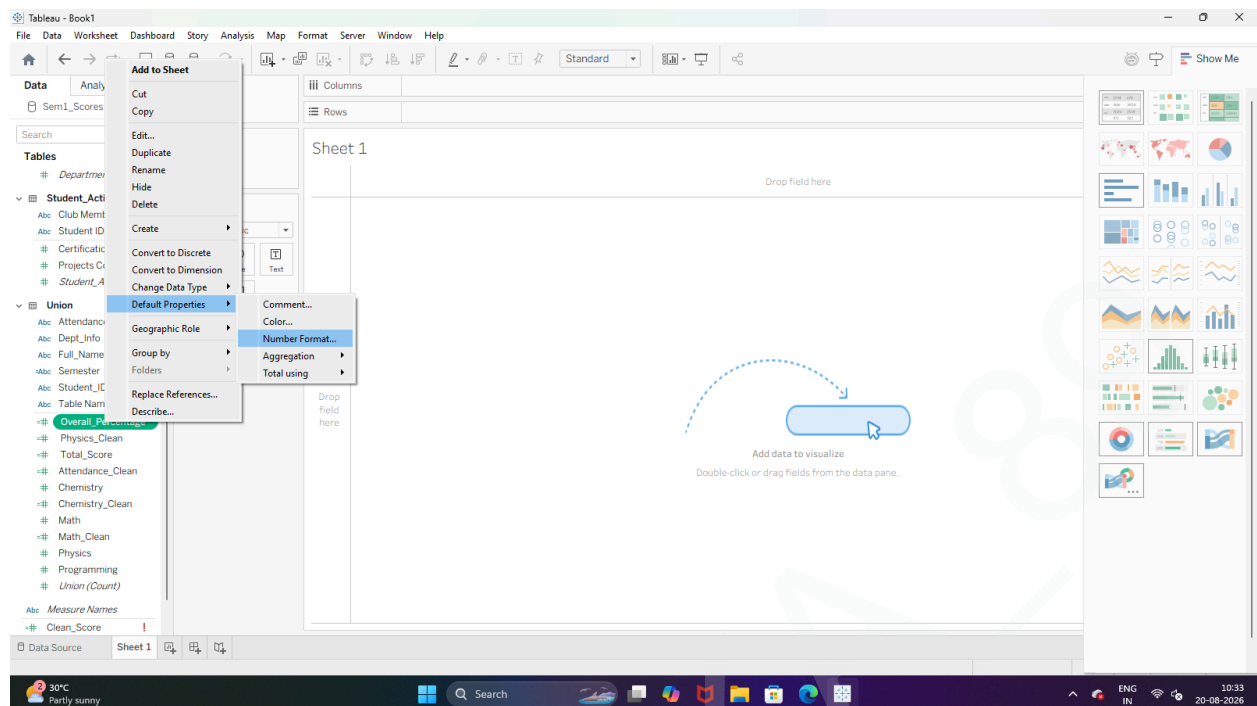
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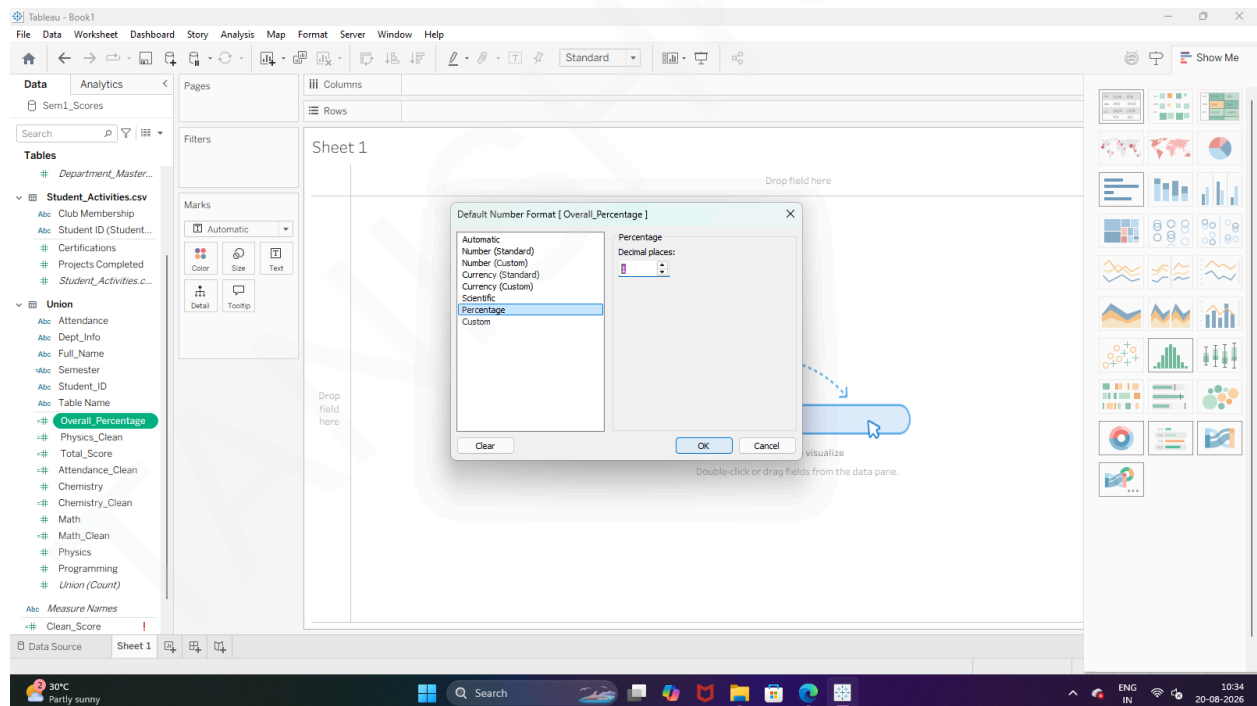
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o Select Percentage and set decimal places to 1 (e.g., 85.5%). o Click OK.



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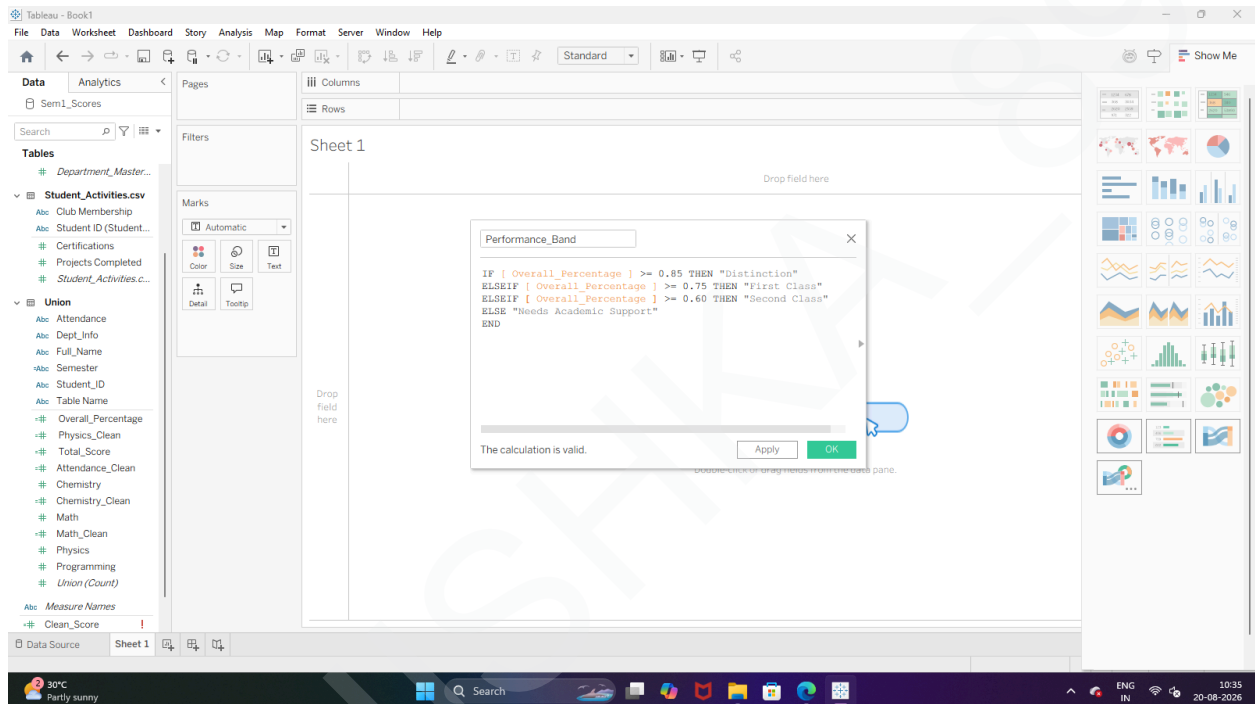
Step 2: Create Categorical Performance Bands (Grading Tiers)

1. Right-click in the Data Pane -->select Create Calculated Field....

2. Create the academic grading tier field:

o Field Name: Performance_Band

o Tableau Formula: Code snippet IF [Overall_Percentage] >= 0.85 THEN "Distinction" ELSEIF [Overall_Percentage] >= 0.75 THEN "First Class" ELSEIF [Overall_Percentage] >= 0.60 THEN "Second Class" ELSE "Needs Academic Support" END o Click OK.



Step 3: Create an Attendance Health Flag

1. Right-click in the Data Pane -->select Create Calculated Field....

2. Create the attendance compliance indicator:

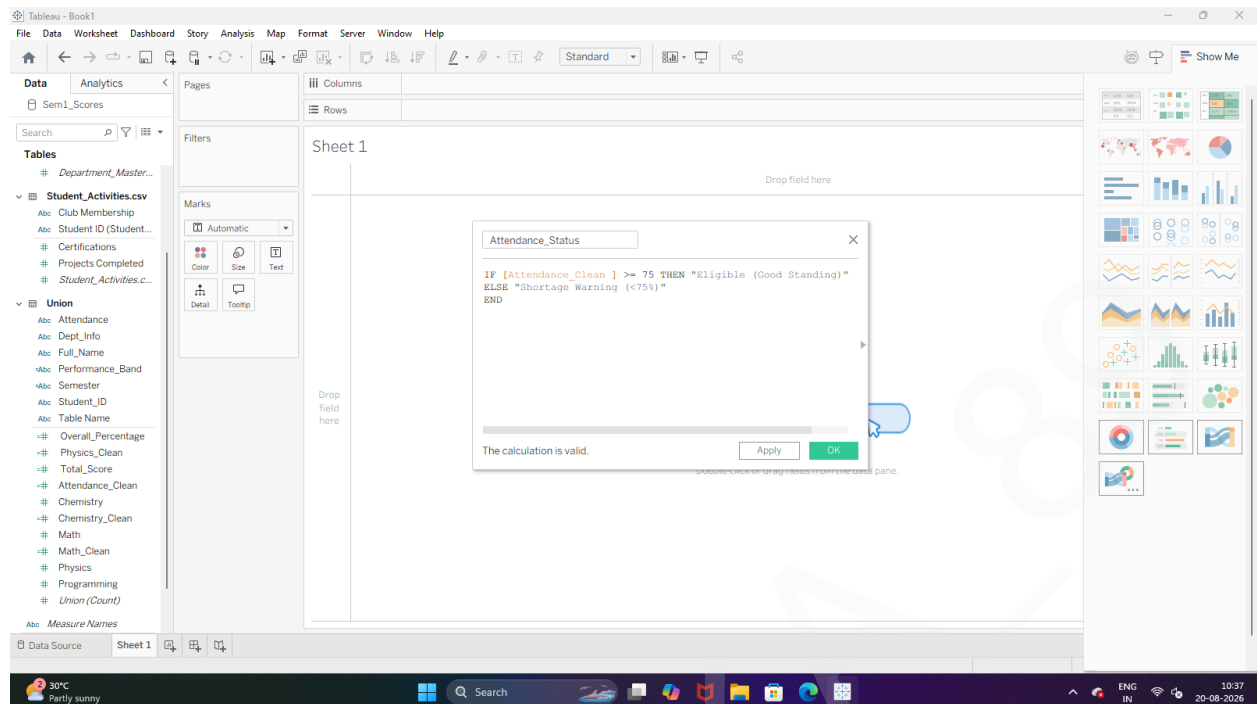
o Field Name: Attendance_Status o Tableau Formula: Code snippet IF [Attendance_Clean] >= 75 THEN "Eligible (Good Standing)" ELSE "Shortage Warning (<75%)" END o Click OK.

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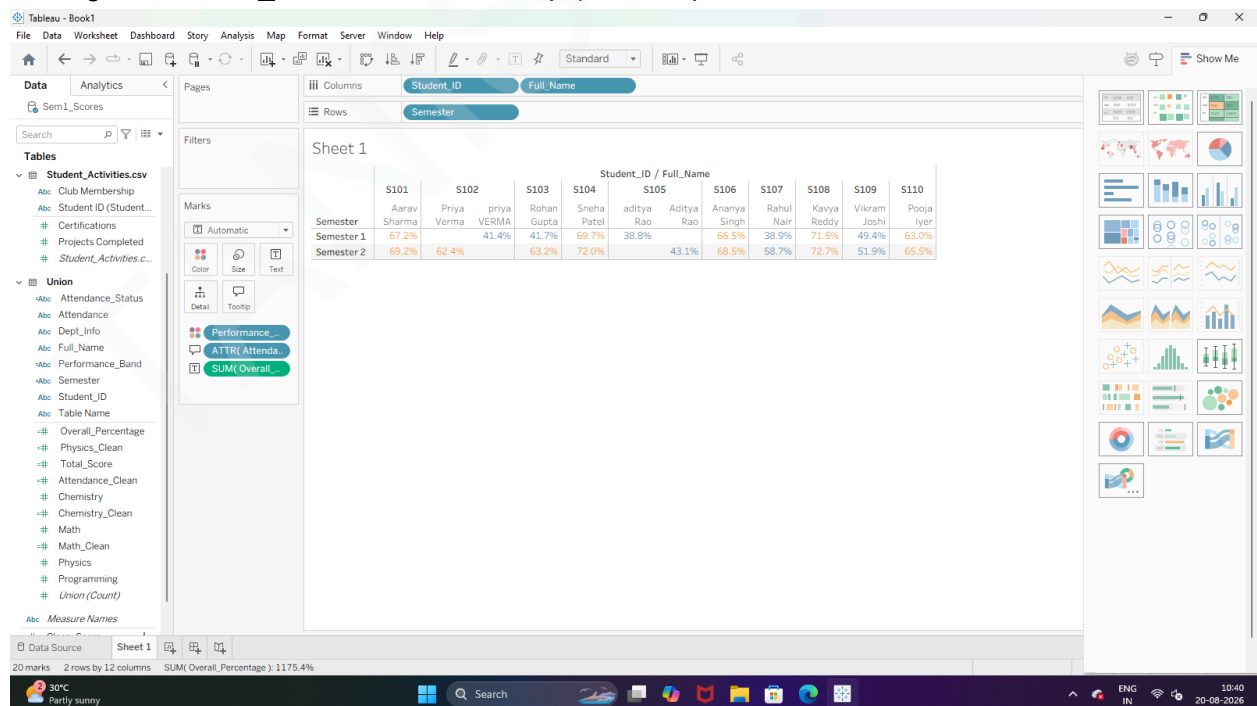
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Step 4: Build the Student Performance Scorecard View

1. Drag Student_ID and Full_Name onto the columns shelf.
2. Drag Semester onto the Rows shelf.
3. Drag Overall_Percentage onto the Text tile on the Marks card.
4. Drag Performance_Band onto the Color tile on the Marks card.
5. Drag Attendance_Status onto the Tooltip (or Detail) tile on the Marks card.



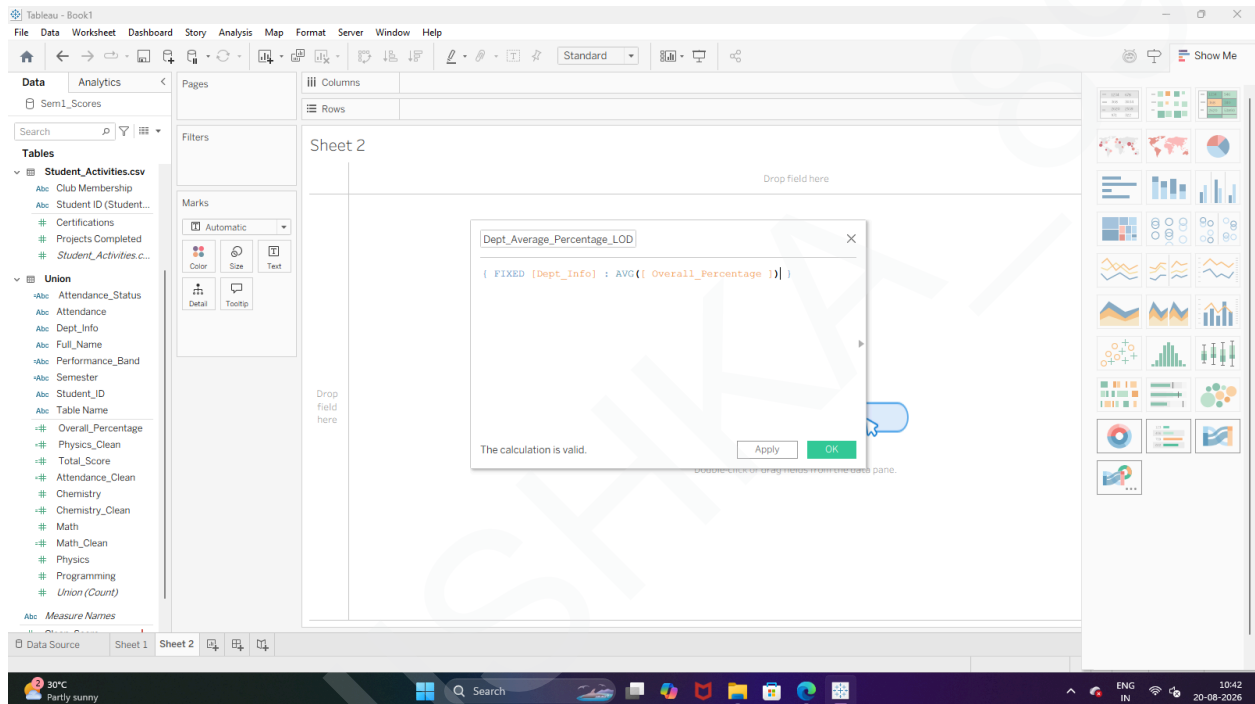
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F: Handle Different Levels of Detail in Datasets

Step 1: Compute Department-Level Benchmark using {FIXED} LOD

1. In Tableau, navigate to Sheet 1 (or create a new worksheet named Sheet 2).
2. In the Data Pane on the left, right-click an empty space --> select Create Calculated Field....
3. Create the department-wide benchmark field:
 - o Field Name: Dept_Average_Percentage_LOD
 - o Tableau Formula: Code snippet { FIXED [Dept_Info] : AVG([Overall_Percentage]) }
 - o Click OK.



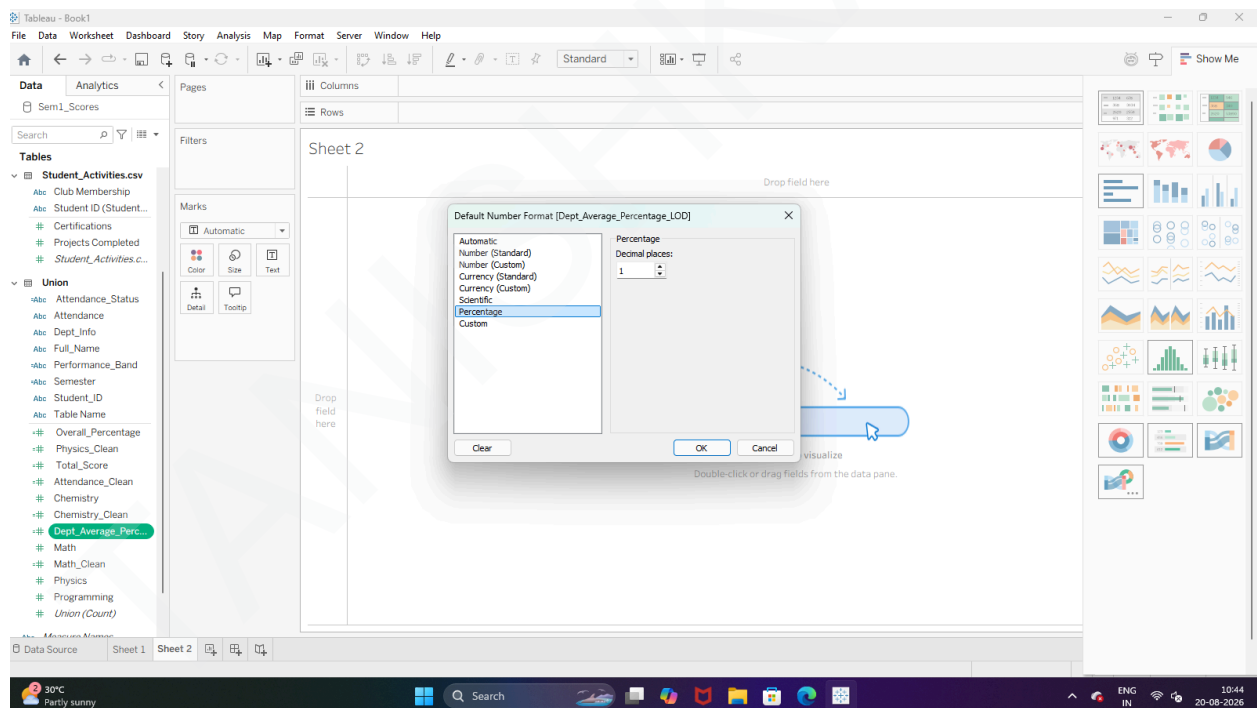
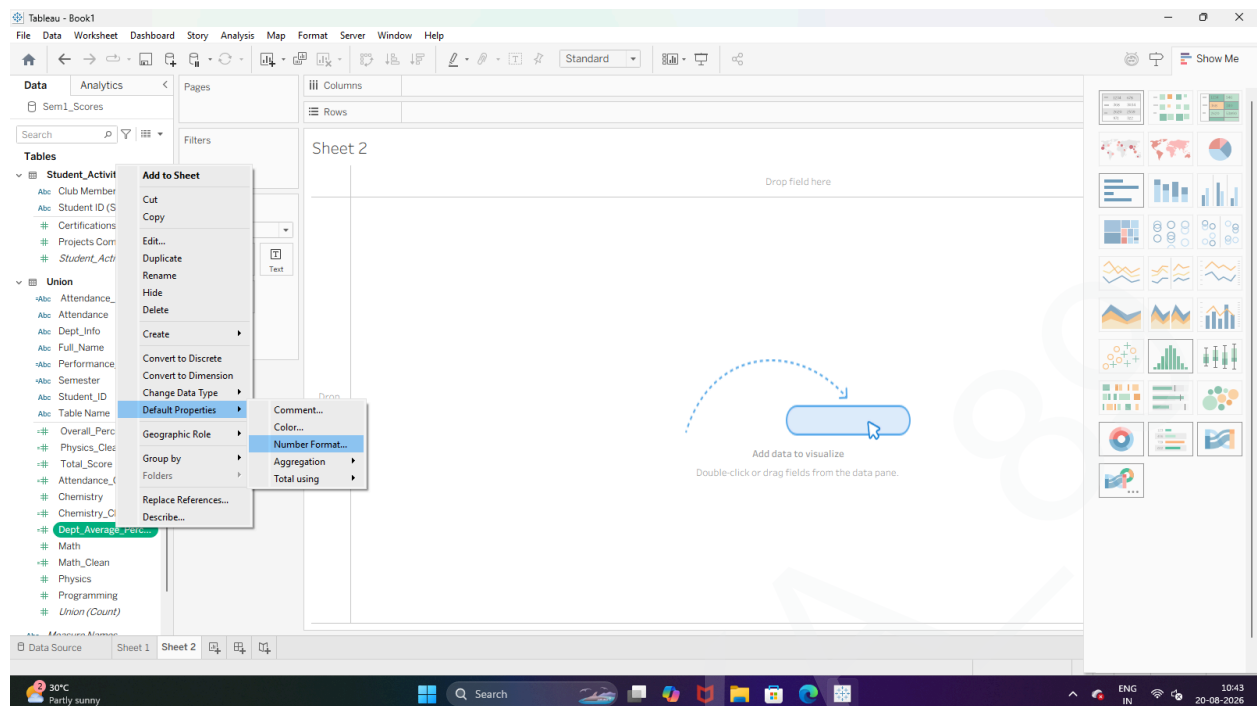
4. Set default formatting: o Right-click Dept_Average_Percentage_LOD --> Default Properties --> Number Format... --> select Percentage (1 decimal place) --> OK.

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Step 2: Compute Student Multi-Semester Average using {FIXED} LOD

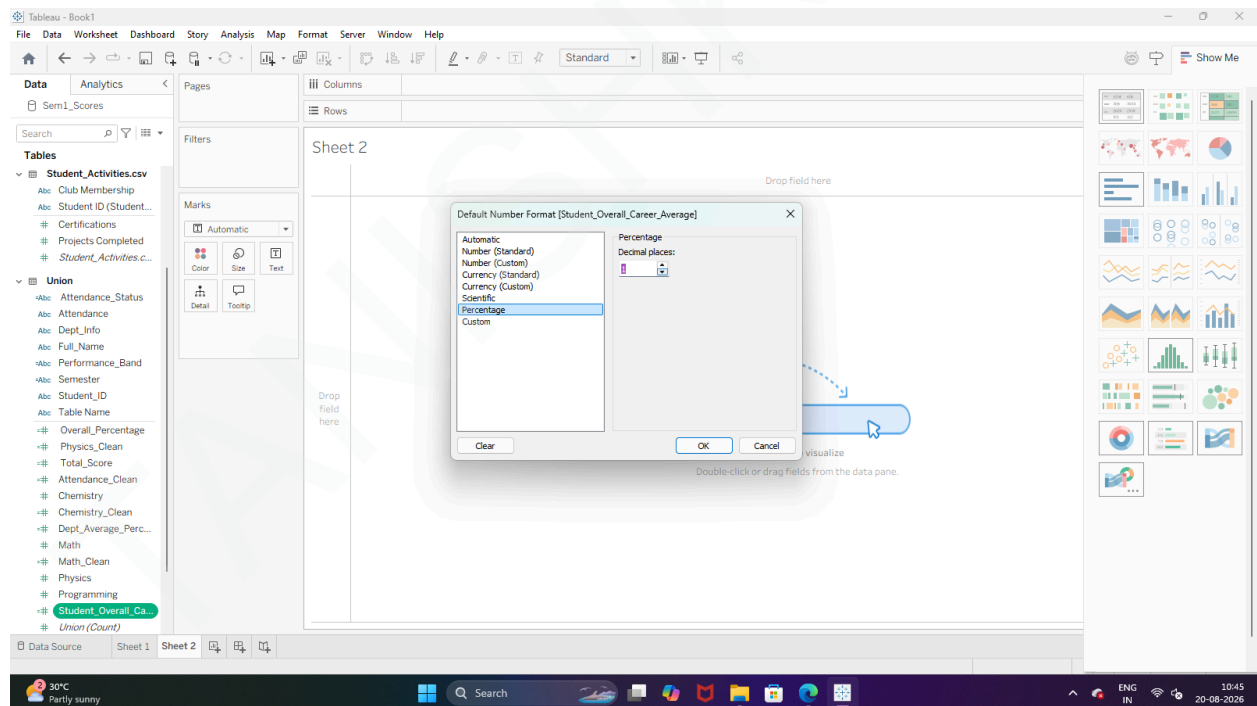
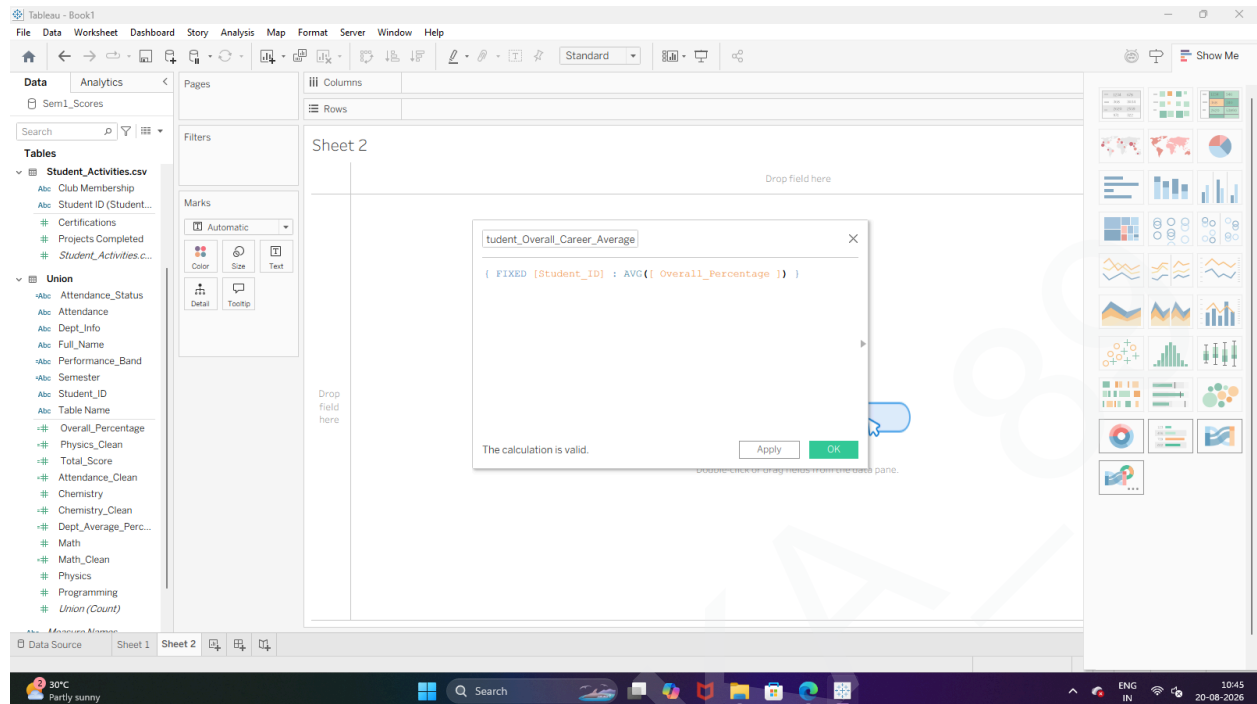
1. Right-click in the Data Pane --> select Create Calculated Field....
2. Create the cumulative student benchmark across all semesters:
 - o Field Name: Student_Overall_Career_Average o Tableau Formula: Code snippet { FIXED [Student_ID] : AVG([Overall_Percentage]) }
 - o Click OK.

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3. Set its Number Format to Percentage (1 decimal place, Just like we did in previous step).



Step 3: Calculate Performance Variance (Student vs. Department Benchmark)

1. Right-click in the Data Pane --> select Create Calculated Field....

2. Create the relative difference metric:

o Field Name: Variance_From_Dept_Average o Tableau Formula: Code snippet
[Overall_Percentage] - [Dept_Average_Percentage_LOD]

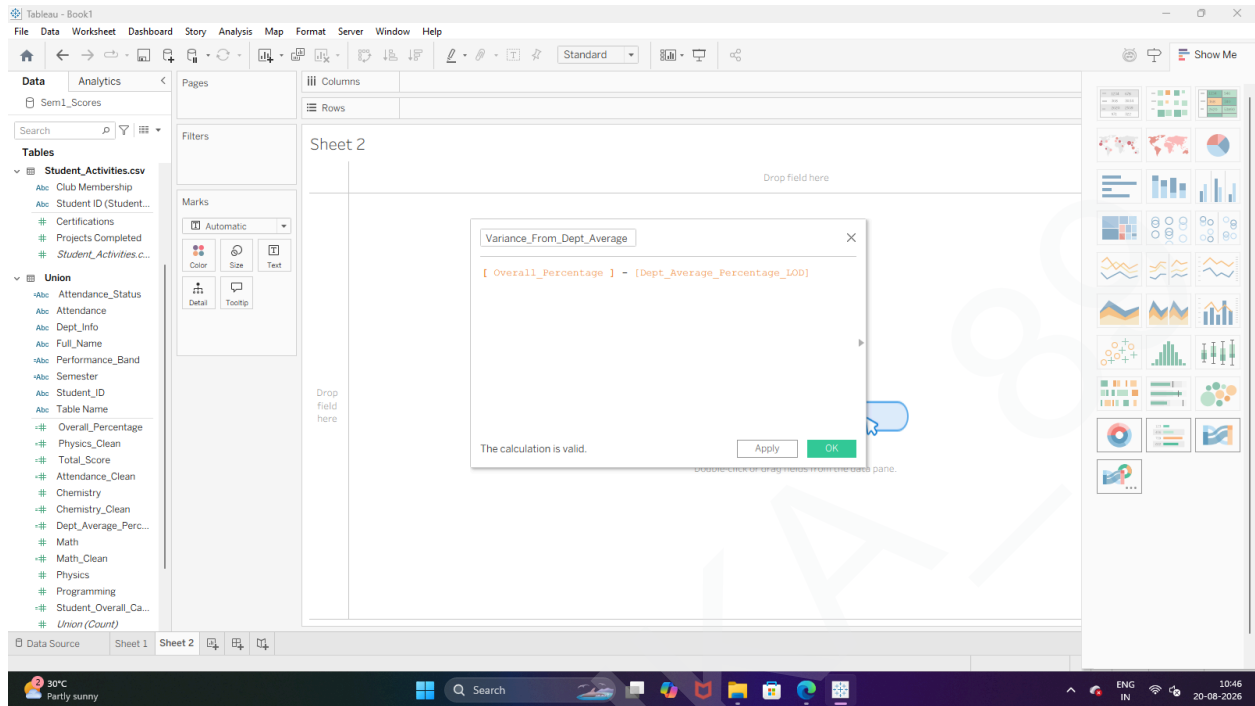
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o Click OK.

3. Set its Number Format to Percentage (1 decimal place).



Step 4: Create a Benchmark Indicator Dimension

1. Right-click in the Data Pane -->select Create Calculated Field....

2. Create an automated comparison label: o Field Name: Benchmark_Status o Tableau

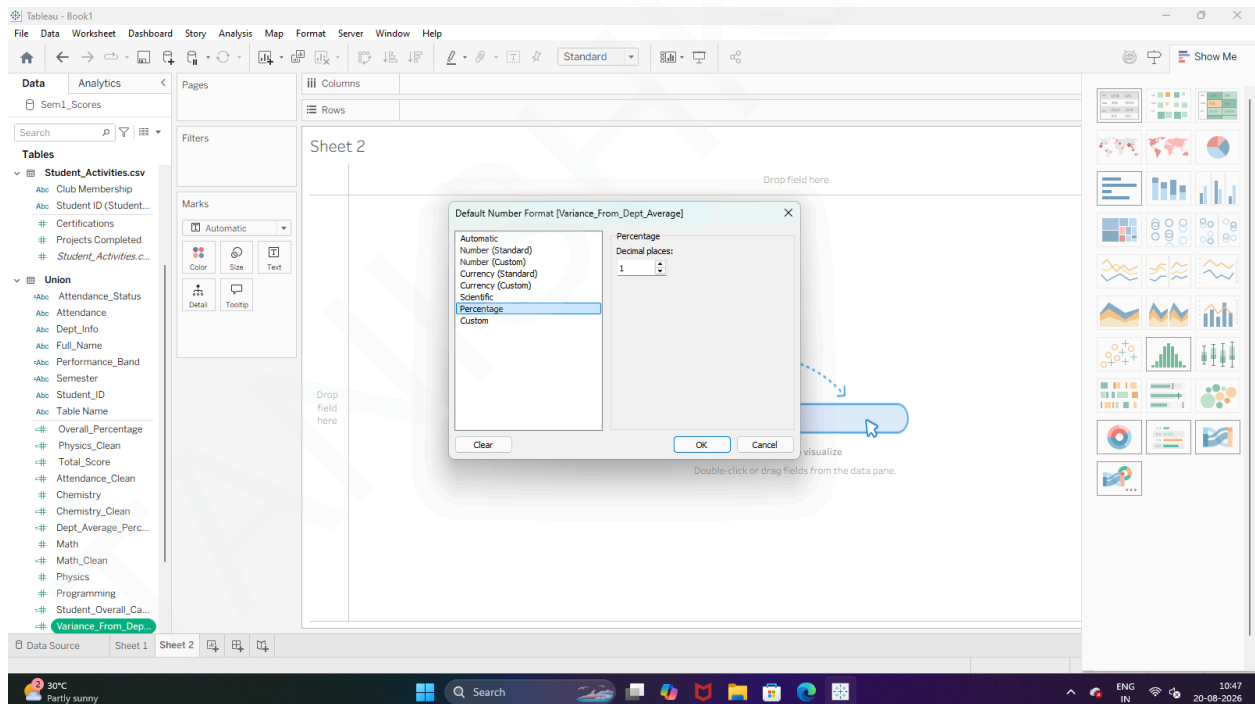
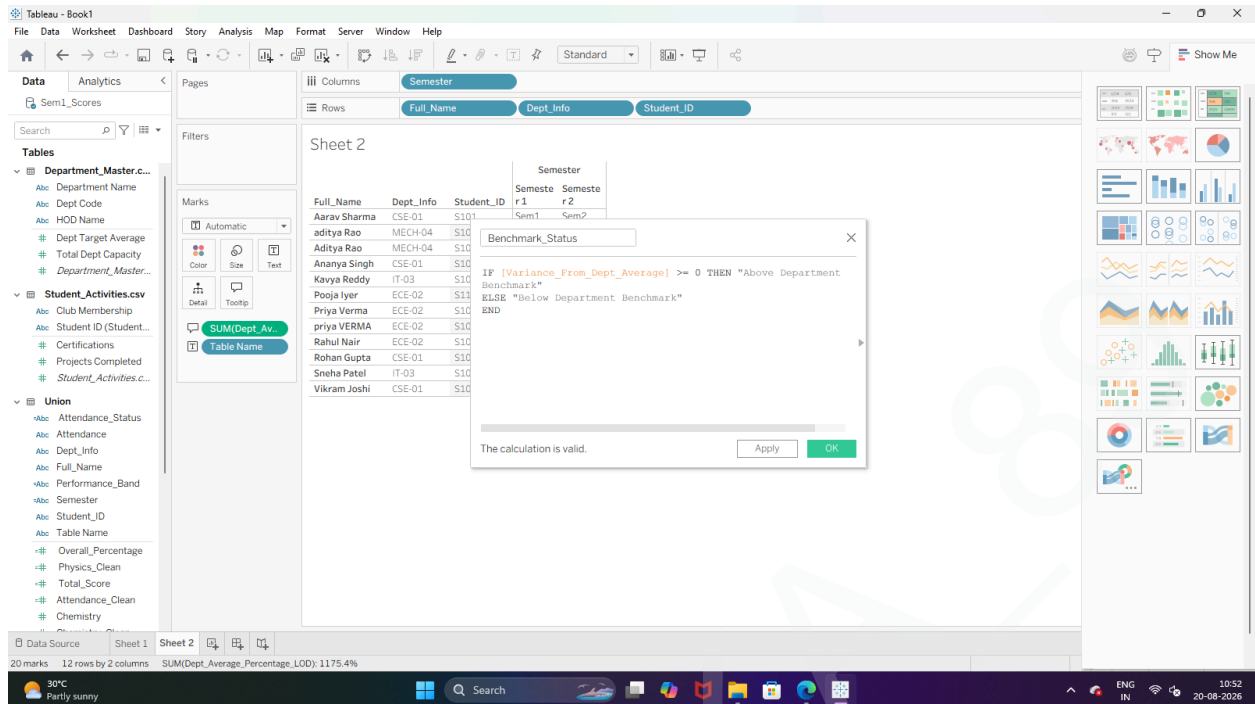
Formula: Code snippet IF [Variance_From_Dept_Average] >= 0 THEN "Above Department Benchmark" ELSE "Below Department Benchmark" END o Click OK.

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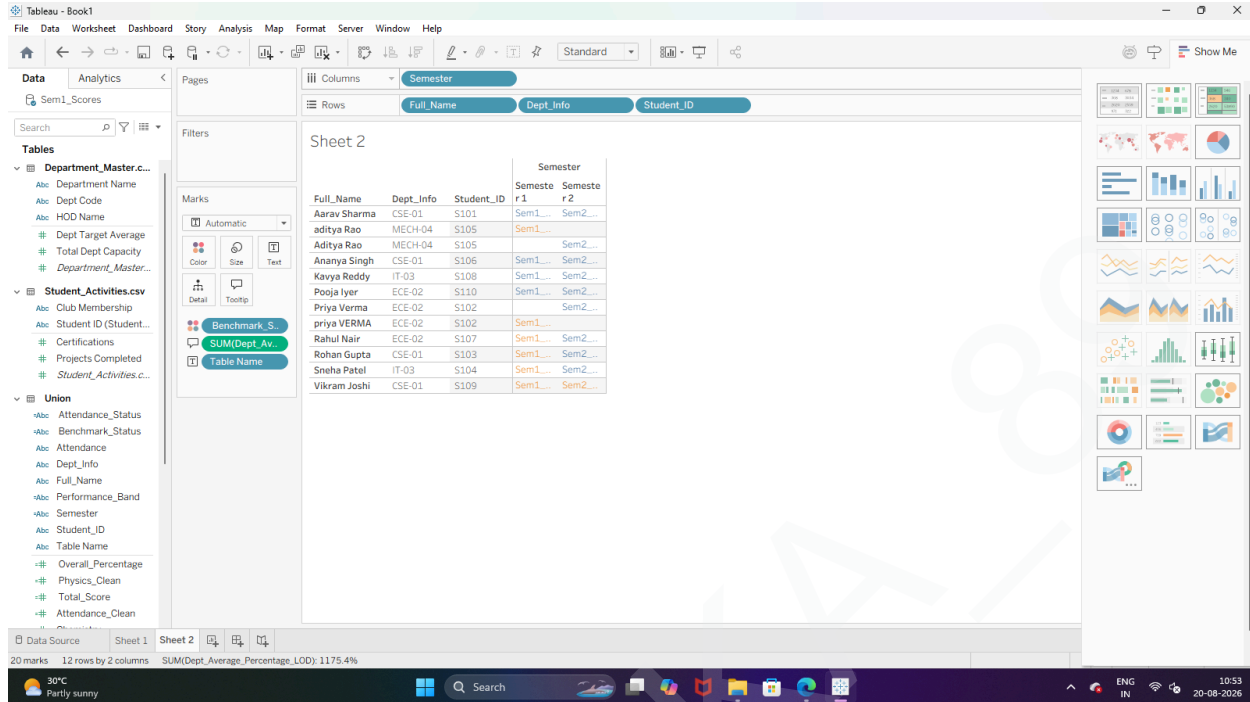
Step 5: Build the LOD Benchmark Visualization

1. Drag Dept_Info, Student_ID, and Full_Name to the Rows shelf.
2. Drag Semester to the Columns shelf.
3. Drag Overall_Percentage onto Text on the Marks card.
4. Drag Dept_Average_Percentage_LOD onto Tooltip on the Marks card.
5. Drag Benchmark_Status onto Color on the Marks card.

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G: Apply Advanced Data Cleaning Techniques

Step 1: Clean Leading/Trailing Whitespace and Inconsistent Name Casing

1. Navigate to Sheet 1 (or a new worksheet). 2. Right-click an empty space in the Data Pane --> select Create Calculated Field....

3. Create the trimmed and standardized student name field:

o Field Name: Full_Name_Clean o Tableau Formula: Code snippet

UPPER(LEFT(TRIM([Full_Name]), 1)) + LOWER(MID(TRIM([Full_Name]),
2, FIND(TRIM([Full_Name]), " ") - 1)) + " " + UPPER(MID(TRIM([Full_Name]),
FIND(TRIM([Full_Name]), " ") + 1, 1)) + LOWER(MID(TRIM([Full_Name]),
FIND(TRIM([Full_Name]), " ") + 2, LEN(TRIM([Full_Name]))))

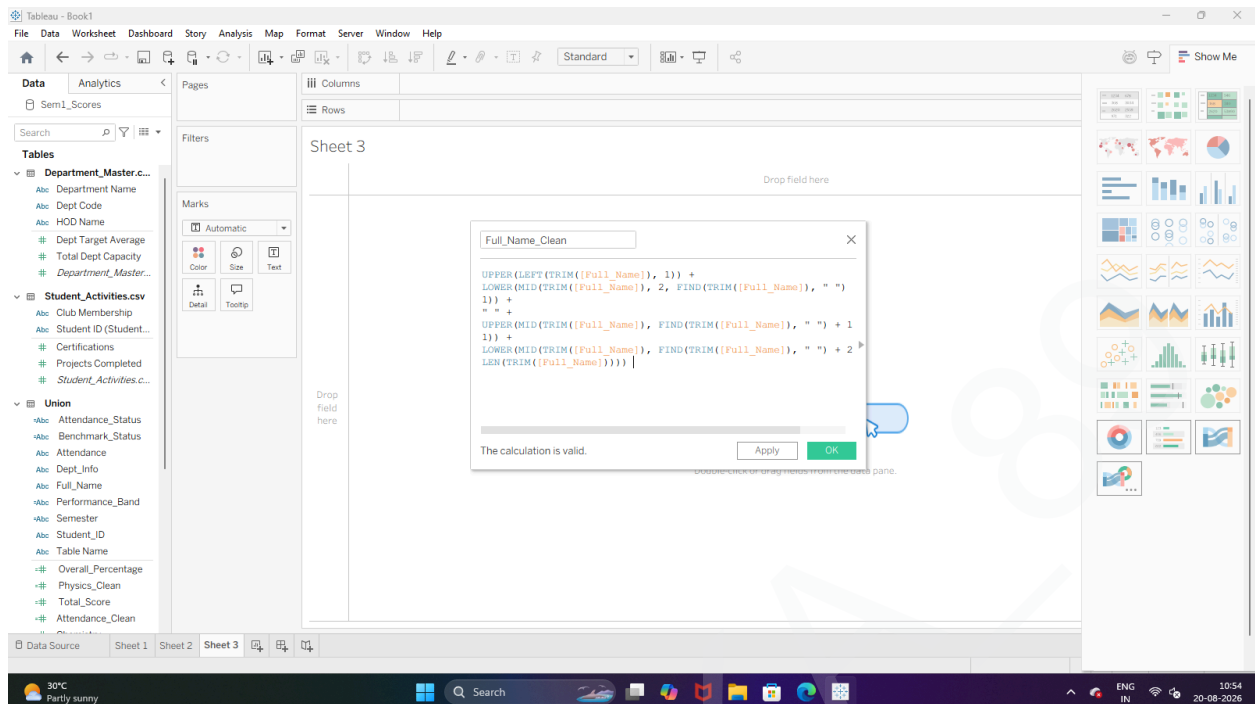
o (Alternatively, for standard trim-only cleaning): Code snippet TRIM([Full_Name])

o Click OK.

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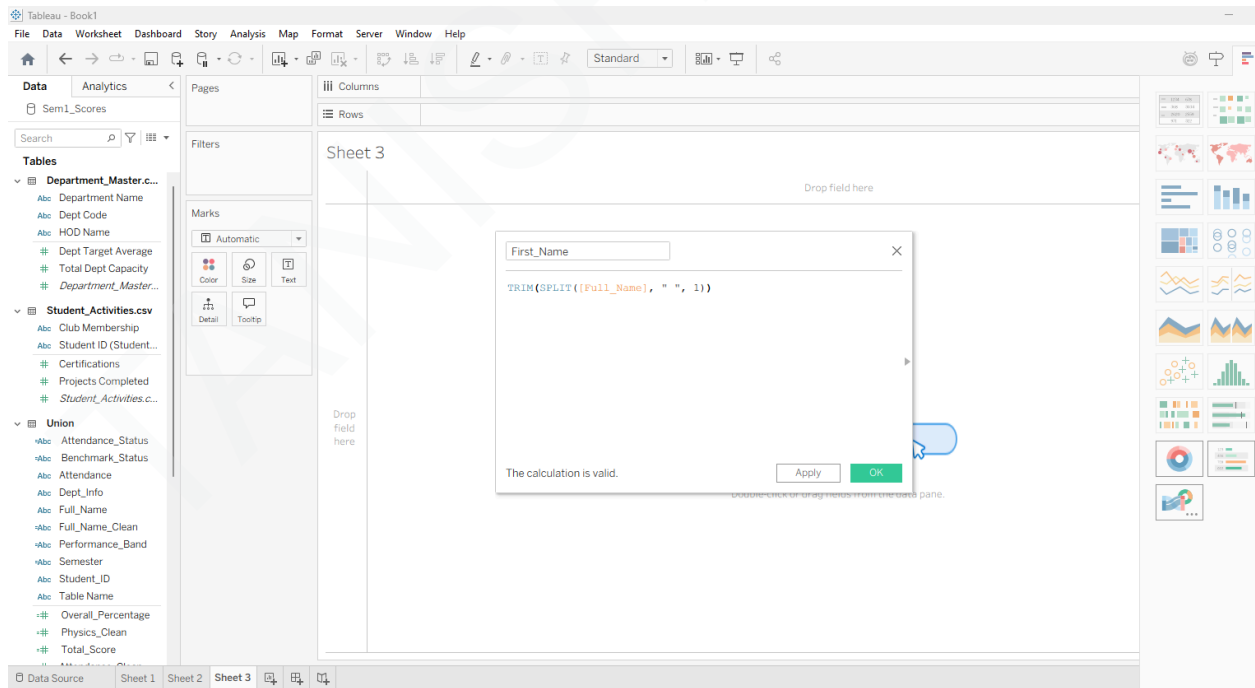
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Step 2: Parse First Name and Last Name using SPLIT()

1. Extract the First Name: o Right-click in the Data Pane --> select Create Calculated Field....
o Field Name: First_Name o Tableau Formula: Code snippet TRIM(SPLIT([Full_Name], " ", 1))
o Click OK.



2. Extract the Last Name: o Right-click in the Data Pane --> select Create Calculated Field....

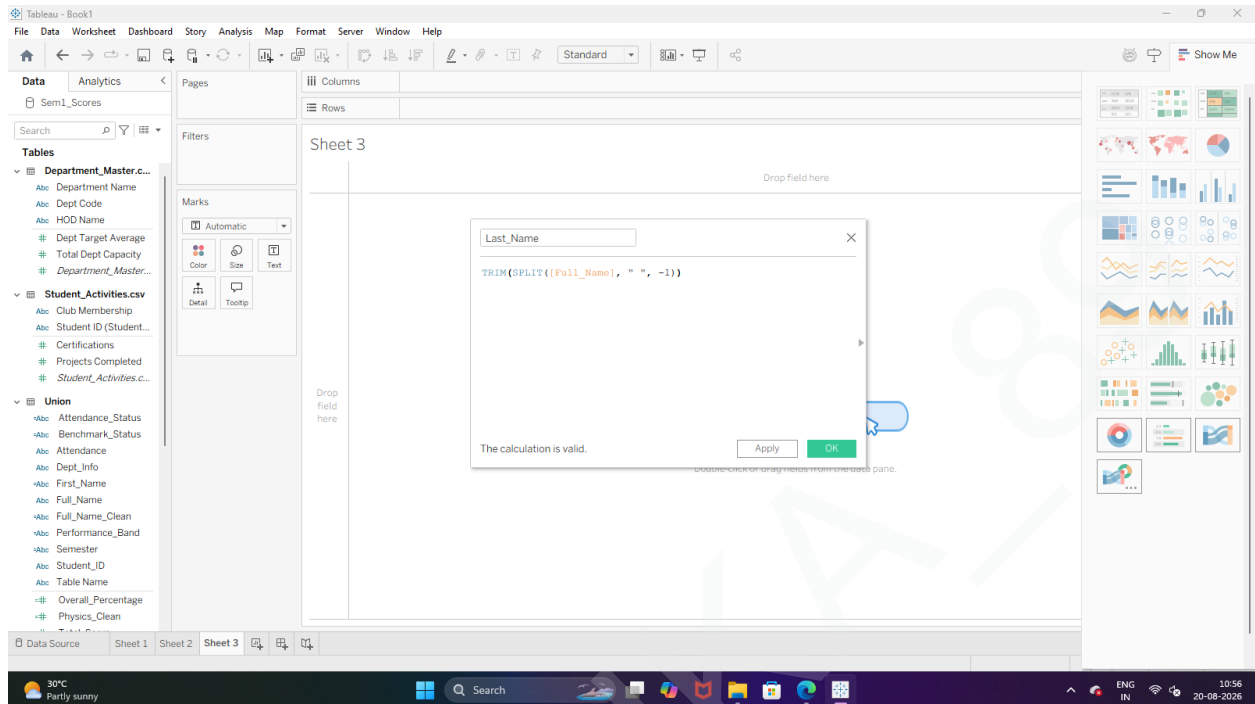
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- o Field Name: Last_Name
- o Tableau Formula: Code snippet `TRIM(SPLIT([Full_Name], " ", -1))`
- o Click OK.



Step 3: Split Composite Department Codes (Dept_Info) The column Dept_Info contains combined strings like CSE-01, ECE-02, and MECH-04. Separate them into distinct dimensional attributes:

1. Extract the Department Acronym:

- o Right-click in the Data Pane --> select Create Calculated Field....
- o Field Name: Dept_Branch
- o Tableau Formula: Code snippet `SPLIT([Dept_Info], "-", 1)`
- o Click OK.

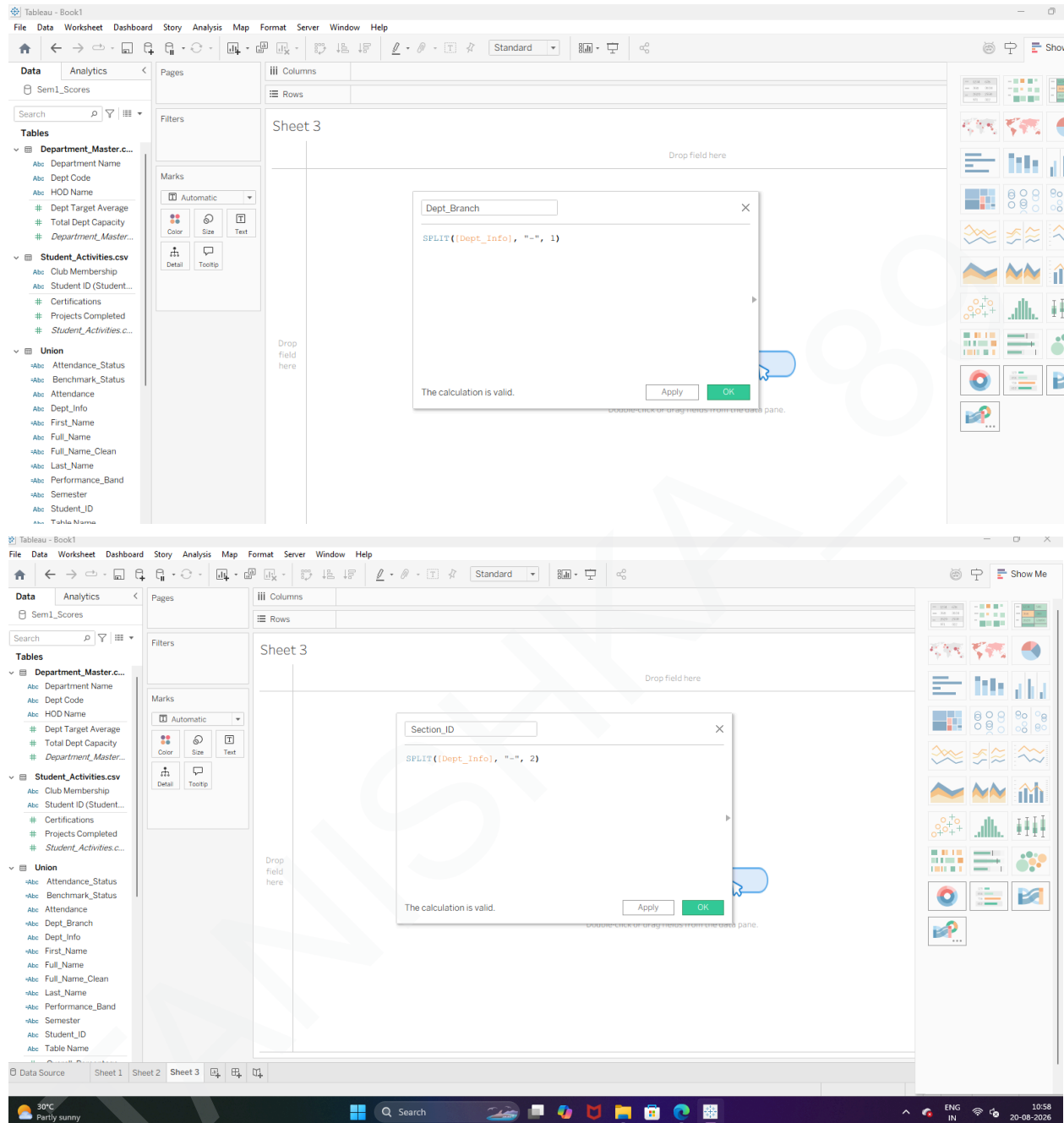
2. Extract the Batch / Section ID:

- o Right-click in the Data Pane --> select Create Calculated Field....
- o Field Name: Section_ID
- o Tableau Formula: Code snippet `SPLIT([Dept_Info], "-", 2)`

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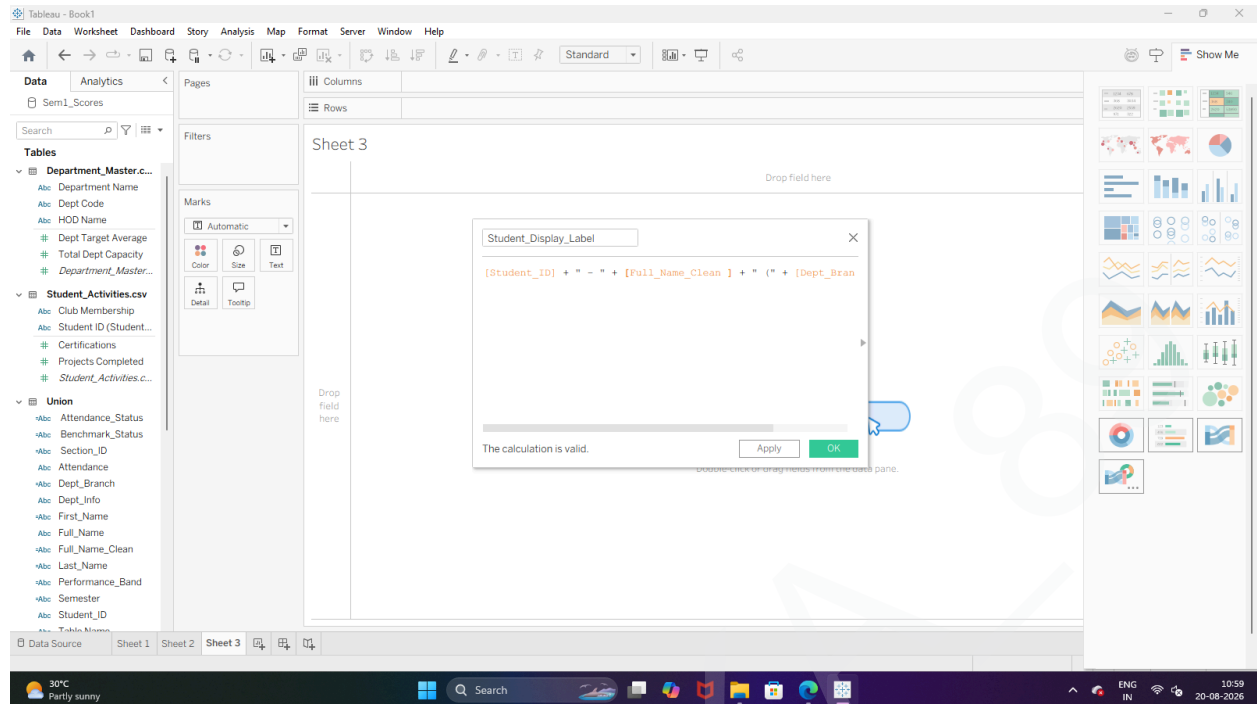


Step 4: Create a Clean Composite Student Label Combine cleaned names, student IDs, and department codes for clean dashboard titles and search filters:

1. Right-click in the Data Pane --> select Create Calculated Field....
2. Field Name: Student_Display_Label
3. Tableau Formula: Code snippet [Student_ID] + " - " + [Full_Name_Clean] + " (" + [Dept_Branch] + ")"
4. Click OK.

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Step 5: Verify in the Final Dashboard View

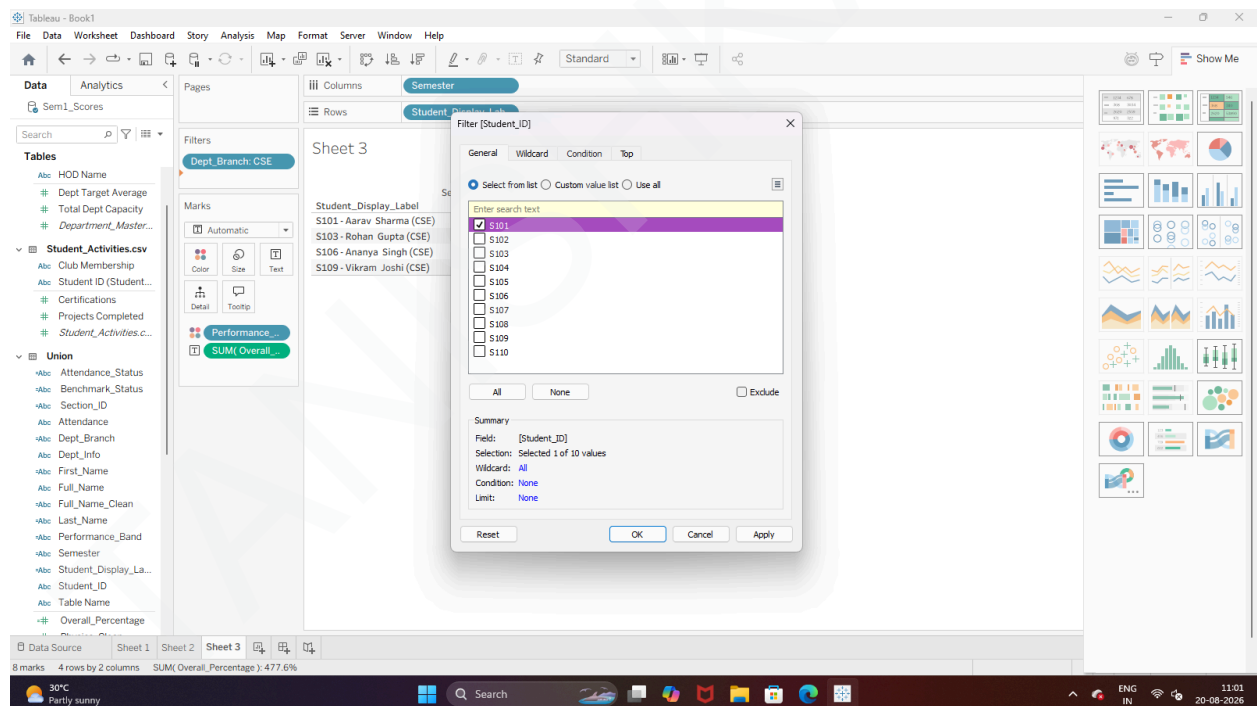
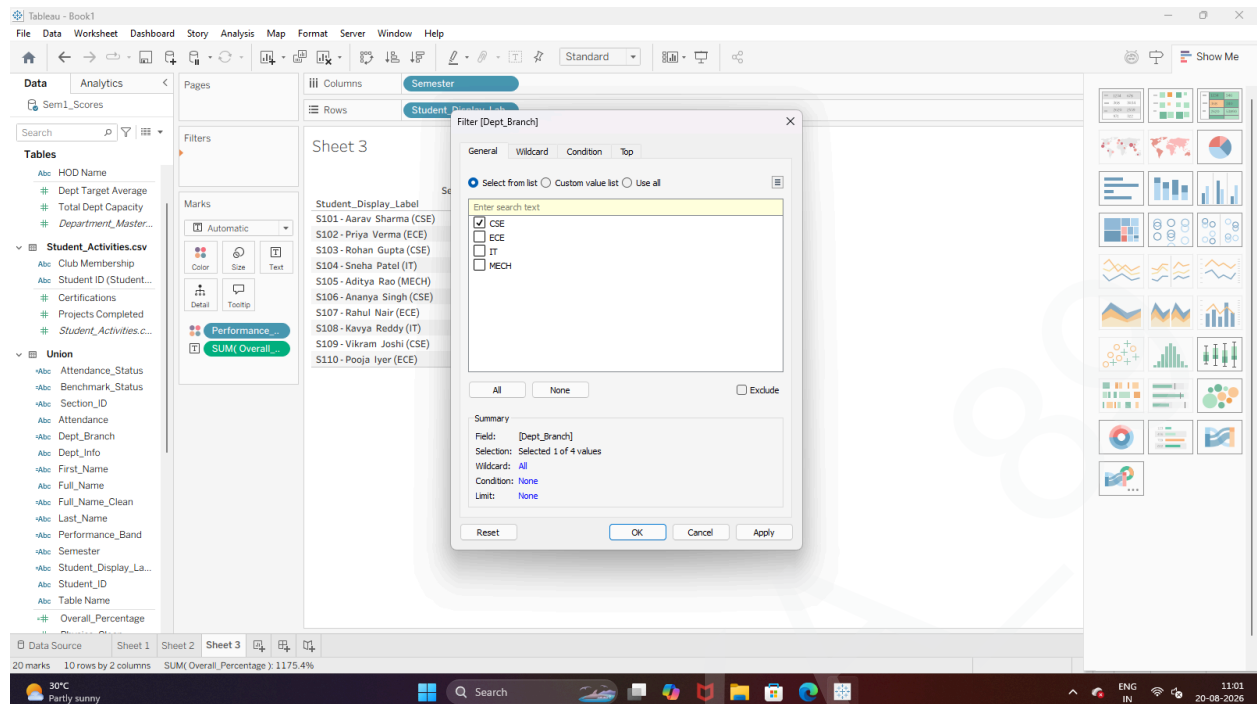
1. Place Student_Display_Label on the Rows shelf.
2. Place Semester on the Columns shelf.
3. Place Overall_Percentage on Text.
4. Place Performance_Band on Color.
5. Place Dept_Branch and Section_ID in the Filters shelf. (While Filtering You Can select any of the value you want)

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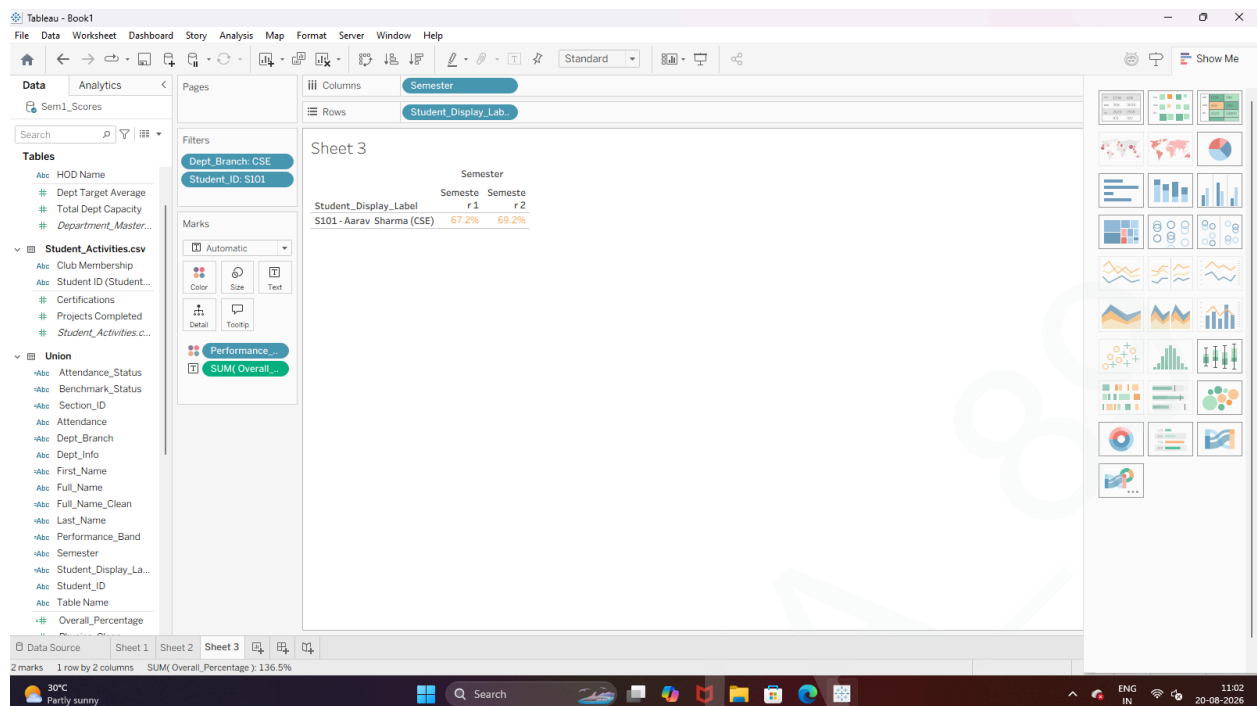


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